

Slovak University of Technology  
Faculty of Informatics and Information Technologies

FIIT-182905-116308

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**E-Learning SW modeling using a Class Diagram**

Master's Thesis

Supervisor: Ing. Vladimír Mlynarovič, PhD.

May 2026



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Master's Thesis

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| Study programme:    | Intelligent Software Systems                                              |
| Study field:        | Informatics                                                               |
| Training workplace: | Institute of Informatics,<br>Information Systems and Software Engineering |
| Supervisor:         | Ing. Vladimír Mlynarovič, PhD.                                            |
| May 2026            |                                                                           |



I declare in my honor that I have prepared this work independently, on the basis of consultations and using the mentioned literature.

In Bratislava, May 2026

Bc. Matej Ševčík



## **Annotation**

Slovak University of Technology Bratislava  
Faculty of Informatics and Information Technologies  
Degree Course: Intelligent Software Systems

Author: Bc. Matej Ševčík  
Diploma Thesis: E-Learning SW modeling using a Class Diagram  
Supervisor: Ing. Vladimír Mlynarovič, PhD.  
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E-learning is a perspective field that offers many opportunities to support education. It has the potential to increase the effectiveness of learning, reduce the burden on educators, make education more accessible to the general public, and introduce innovative practices that traditional educational approaches do not allow. Despite this, e-learning is used relatively rarely in Slovak schools, even at faculties of informatics. In this project, we focused on the possibilities of applying e-learning to teaching the UML class diagram, with the potential for expansion to other UML diagrams. We analyzed various techniques used in e-learning and their application in teaching software modeling. The main goal of this project is to create a course for teaching the UML class diagrams. Based on our analysis, we selected several gamification e-learning practices and implemented them in the course using two different content designs - competitive and playful. We conducted an experiment on students and compared the effects of the designs on their test results and subjective perceptions. The main contribution of the project is that our course, along with the analysis and tests, has created a foundation for further research and improvement of the application of e-learning in the field of software modeling at Faculty of Informatics and Information Technologies STU in Bratislava (FIIT STU).

Keywords: e-learning, class diagram, software modeling





## Anotácia

Slovenská technická univerzita v Bratislave  
Fakulta informatiky a informačných technológií  
Študijný program: Intelligent Software Systems

Autor: Bc. Matej Ševčík

Diplomová práca: E-Learning modelovania SW pomocou diagramu tried

Vedúci diplomového projektu: Ing. Vladimír Mlynarovič, PhD.

May 2026

E-learning je rýchlo sa rozvíjajúca oblasť poskytujúca veľa možností podpory vzdelávania. Má potenciál zvýšiť efektivitu vzdelávania, znížiť záťaž na pedagógov, sprístupniť vzdelávanie širšej verejnosti alebo zaviesť inovatívne praktiky, ktoré tradičné vzdelávacie prístupy neumožňujú. Napriek tomu je na slovenských školách e-learning využívaný pomerne zriedkavo, a to aj na fakulte informatiky. V projekte sme sa zamerali na možnosti aplikovania e-learningu pri vyučovaní UML diagramu tried s možnosťou rozšírenia na ďalšie UML diagramy. Vykonali sme analýzu rôznych prístupov používaných v e-learningu a ich využitia pri vyučovaní modelovania softvéru. Hlavným cieľom tohto projektu je vytvorenie kurzu na výučbu UML diagramu tried, v ktorom sme na základe implementovali niekoľko vybraných e-learningových praktík. Použili sme pri tom dva prístupy ku dizajnu kurzu - súťaživý a hravý. Vykonali sme experiment na študentoch, v ktorom sme porovnali vplyv týchto dizajnov na ich objektívne výsledky a subjektívne pocity. Prínos projektu spočíva najmä v tom, že náš kurz, analýza a testy pripravili priestor na ďalšie rozširovanie a vylepšovanie možností e-learningu v oblasti modelovania softvéru na acfiit.

Kľúčové slová: e-learning, diagram tried, modelovanie softvéru



# Návrh zadania diplomovej práce

*Finálna verzia do diplomovej práce<sup>1</sup>*

## Študent:

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## Výskumník:

**Meno, priezvisko, tituly:** Vladimír Mlynarovič, Ing. PhD.

## Projekt:

**Názov:** E-Learning modelovania SW pomocou diagramu tried  
**Názov v angličtine:** E-Learning SW modeling using a class diagram  
**Miesto vypracovania:** Ústav informatiky, informačných systémov a softvérového inžinierstva, FIIT STU  
**Oblasť problematiky:** e-learning, diagram tried, modelovanie softvéru

## Text návrhu zadania<sup>2</sup>

Diagram tried je najpoužívanejším UML diagramom. Slúži na grafické zobrazenie štruktúry systému pomocou tried, ich atribútov, metód a vzťahov medzi nimi. Tento diagram je významným nástrojom pri analýze, návrhu a dokumentácii softvérových systémov. E-learning je rýchlo sa rozvíjajúca oblasť poskytujúca veľa možností podpory vzdelávania. Má potenciál zvýšiť efektivitu vzdelávania, znížiť záťaž na pedagógov alebo sprístupniť vzdelávanie širšej verejnosti. V štúdiách [1][2] študenti, ktorí sa učili modelovať diagram tried pomocou navrhovaného e-learning modulu, dosiahli v cvičeniach a neskôr aj v testoch lepšie výsledky než skupina, ktorá strávila rovnaký čas štúdiom diagramu tried z iných zdrojov. To demonštruje potenciál e-learningu zvýšiť kvalitu výučby v tejto oblasti. Výskumným problémom je teda optimalizácia výučby modelovania softvéru s využitím e-learningových prístupov, pričom hlavnou otázkou je, ktoré metódy sú najefektívnejšie a ako ich vhodne kombinovať.

Analyzujte existujúce e-learningové prístupy vo vzdelávaní softvérového inžinierstva so zameraním na výučbu UML diagramov. Navrhnete vzdelávací nástroj, ktorý implementuje poznatky z tejto analýzy a sprístupníte ho v rámci prototypu novej e-learningovej platformy, prípadne ako modul do už existujúcej platformy. Otestujte efektivitu nástroja na študentoch bakalárskeho štúdia na FIIT STU pomocou testov. Porovnajte výsledky študentov, ktorí využili na prípravu nami navrhnutý nástroj a tých, ktorí sa učili z iných zdrojov. Prínos projektu bude spočívať najmä v tom, že náš nástroj, analýza a testy pripravia priestor na ďalšie rozširovanie a vylepšovanie možností e-learningu v oblasti modelovania softvéru.

<sup>1</sup> Vytlačiť obojstranne na jeden list papiera

<sup>2</sup> 150-200 slov (1200-1700 znakov), ktoré opisujú výskumný problém v kontexte súčasného stavu vrátane motivácie a smerov riešenia



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# Chapter 1

## Introduction

Unified Modeling Language (UML) is a widely used modeling language. It is used for describing systems in an abstract way and from various perspectives, which makes the ability to understand it and use it a valuable asset for any professional in the field of information technologies.

Most developers claim to possess knowledge of UML. However, in many cases, their knowledge of the topic is rather limited or even inaccurate<sup>1</sup>. This observation highlights the need for improving the approach to teaching of software modeling and UML in educational institutions.

Traditional classes often do not provide enough time and space for an actual teacher to give the necessary amount of attention to each of the students. The answer to this problem might lie in electronic learning, or e-learning. It is a practice in which learning material is delivered through digital technologies.

E-learning can offer students interactive materials with automated immediate personalized feedback, structured and easily accessible learning materials or study and exercise recommendations, while simultaneously allowing teachers to spend more time consulting students instead of explaining the basic concepts to them.

E-learning also provides possibilities for increasing students' engagement in the study materials by introducing gamification elements or a sense of competition to the courses.

The class diagram is the most prominent diagram in UML. It is also common for students to have trouble using it properly. For these reasons, we decided to aim this study at designing an e-learning course that would act as complementary material to a software

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<sup>1</sup>Donald Bell, "An Introduction to the Unified Modeling Language," *IBM Developer*, <https://developer.ibm.com/articles/an-introduction-to-uml/>, accessed February 23, 2025.

modeling university subject.

Our course will implement a number of modern e-learning approaches to improve information retention, engagement, and the overall learning experience of the students. It will be tested on a sample of students at FIIT STU. The results will then be analyzed to determine the effectiveness and further potential of such tools in the process of education of software modeling.





## Chapter 2

# The Class Diagram

### 2.1 Overview of UML

#### Introduction

The UML is a standardized visual modeling language. Its first version was published in 1997 and later updated to UML 2.0 in 2005 as an attempt to unify various approaches to modeling software systems

footnoteVisual Paradigm, "What is Unified Modeling Language (UML)?," *Visual Paradigm*, <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-uml/>, accessed February 23, 2025.. The latest version of UML is UML 2.5.1 and it was published in December 2017<sup>1</sup>.

The purpose of the language is to specify, document, and visualize parts of software systems. That is accomplished by designing standardized diagrams, each intended to describe a different aspect of the system, usually from various stakeholders' perspectives. Those may include the architect, programmer, project manager, customer, and others.

UML models contain models of business processes, interactions and relationships between individual components or the structure and functions of components. UML documents all stages of the development of a system, not only the initial part. It is mainly aimed at, but not limited to, object-oriented systems.

A model is a representation of something - in this case, a software system. A model captures the system from a certain viewpoint and omits its aspects that are not relevant to the viewpoint [1].

---

<sup>1</sup>Conrad Bock et al., "Unified Modeling Language," *Object Management Group (OMG)*, <https://www.omg.org/spec/UML/>, accessed February 23, 2025.

### UML diagrams

UML defines a large number of elements that build up diagrams. There are in total 13 different diagrams defined. Those diagrams represent various aspects and viewpoints of the system. In some sources, 14 diagrams are mentioned. The profile diagram is usually not considered to be a separate diagram type however.

Some diagrams show the structure of the system. Those are called structural diagrams. They do not display the behavior of the modeled elements [2]. They describe the system using objects, attributes, operations and relationships.

The other type of diagrams are behavioral diagrams. They describe how the objects interact and change during runtime. They use sequence, activity, collaboration, and state to represent these changes<sup>2</sup>.

The structural UML diagrams are:

- object diagram
- package diagram
- class diagram
- component diagram
- composition structure diagram
- deployment diagram

The behavioral diagrams are:

- state machine diagram
- use case diagram
- activity diagram
- interaction overview diagram
- sequence diagram
- communication diagram
- timing diagram

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<sup>2</sup>Visual Paradigm, "What is Unified Modeling Language (UML)? (Behavior vs Structural Diagram)," *Visual Paradigm*, <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/behavior-vs-structural-diagram/>, accessed April 13, 2025.



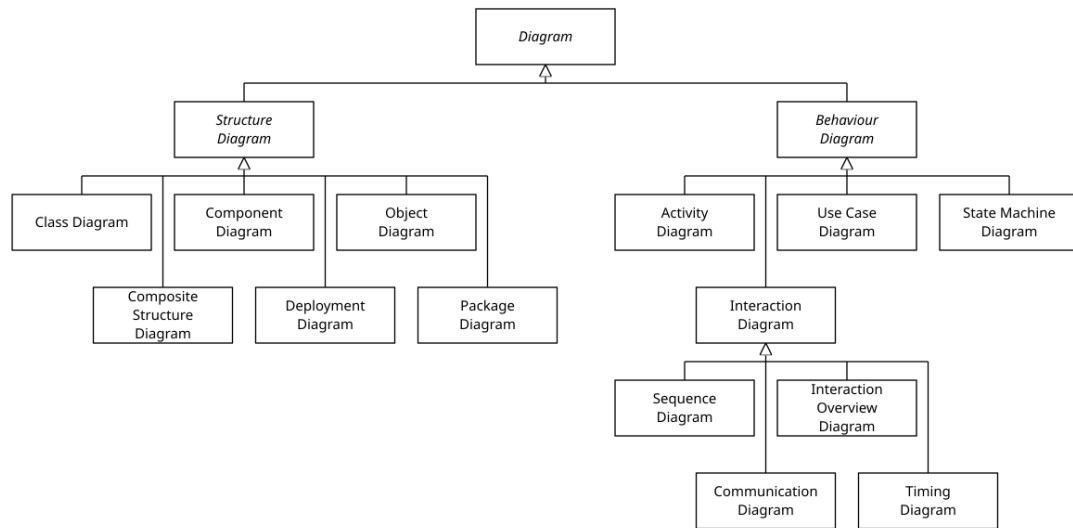


Figure 2.1: Hierarchy of UML diagrams

Source: Dave A. Ryan, "UML diagram," *Wikimedia Commons*, [https://en.m.wikipedia.org/wiki/File:Uml\\_diagram.svg](https://en.m.wikipedia.org/wiki/File:Uml_diagram.svg), Accessed: May 23, 2025.

## 2.2 UML modeling tools<sup>3</sup>

There are many tools for creating UML diagrams available. They differ in complexity, support of features, price or intended use. We selected some of the most popular ones for closer inspection and compared them.

### StarUML<sup>4</sup>

StarUML is a fast, lightweight UML tool geared toward developers. It supports modern UML standards and allows extension via plugins. StarUML is widely appreciated for its clean interface and Markdown-compatible documentation features.

### Enterprise Architect<sup>5</sup>

Enterprise Architect by Sparx Systems is a powerful, professional-grade modeling tool. It supports UML, SysML, BPMN, ArchiMate, and more. EA is often used in large-scale software development and system engineering projects.

### Diagrams.net<sup>6</sup>

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<sup>3</sup>This section was made with the use of Generative artificial intelligence (Gen AI)

<sup>4</sup>"StarUML," *StarUML Official Website*, <https://staruml.io/>, accessed May 24, 2025.

<sup>5</sup>"Enterprise Architect," *Sparx Systems*, <https://sparxsystems.com/products/ea/>, accessed May 24, 2025.

<sup>6</sup>"Diagrams.net (Draw.io)," *Diagrams.net*, <https://www.diagrams.net/>, accessed May 24, 2025.

Diagrams.net, also known as Draw.io, is a free, browser-based diagramming tool. While it's not strictly tailored for UML, it offers sufficient shapes and templates to support simple to moderately complex UML diagrams. It is therefore mainly used for making quick sketches or by students.

### Visual Paradigm<sup>7</sup>

Visual Paradigm is a comprehensive UML and software design tool. It supports all UML diagram types and extends into system modeling, business process modeling, and database design. Visual Paradigm is favored in both academia and industry for its clarity, template variety, and export options. Visual Paradigm also provides a simpler online diagram editing tool that we used for designing diagrams in our study.

### Lucidchart<sup>8</sup>

Lucidchart is an online diagramming application with strong collaboration features. It supports UML and other diagram types like flowcharts, ERDs, and wireframes. It is widely used in business and education settings.

### Overall comparison

| Tool             | StarUML         | Enterprise Architect | Diagrams.net       |
|------------------|-----------------|----------------------|--------------------|
| UML Support      | Full            | Full                 | Partial (manual)   |
| Platform         | Win/macOS/Linux | Windows Desktop      | Web/Desktop        |
| License          | Paid (Low-cost) | Paid (Proprietary)   | Free (Open Source) |
| Collaboration    | No              | Yes                  | Limited            |
| Code Engineering | Yes             | Yes                  | No                 |
| Templates        | Moderate        | Rich                 | Limited            |
| Best Use Case    | Developers      | Enterprise Systems   | Quick sketches     |

Table 2.1: Comparison of UML tools: StarUML, Enterprise Architect, Diagrams.net

| Tool             | Visual Paradigm         | Lucidchart          |
|------------------|-------------------------|---------------------|
| UML Support      | Full                    | Moderate            |
| Platform         | Web/Desktop             | Web                 |
| License          | Free + Paid tiers       | Free + Paid tiers   |
| Collaboration    | Yes                     | Real-time           |
| Code Engineering | Yes                     | No                  |
| Templates        | Rich                    | Rich                |
| Best Use Case    | Academic / Professional | Collaborative Teams |

Table 2.2: Comparison of UML tools: Visual Paradigm and Lucidchart

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<sup>7</sup>"Visual Paradigm," *Visual Paradigm*, <https://www.visual-paradigm.com/>, accessed May 24, 2025.

<sup>8</sup>"Lucidchart," *Lucidchart*, <https://www.lucidchart.com/>, accessed May 24, 2025.

## 2.3 Class Diagram

The class diagram is the most prominent UML diagram. It describes the elements of the system as well as their relationships, creating the basis for numerous other diagrams and modeling techniques [3]. These aspects are static, meaning that they do not change with time. Despite that, it is used in multiple phases of system development, applying various levels of abstraction. It is even possible to generate code from the class diagram. It is commonly used as a part of documentation [2], as it provides an easy to understand overview of the system.

### 2.3.1 Perspectives of the Class Diagram<sup>3</sup>

Class diagrams are one of the most fundamental diagram types in the Unified Modeling Language (UML) and serve various purposes throughout the software development lifecycle. They apply various levels of abstraction and each level of abstraction uses class diagrams slightly differently, but they all help bridge the gap between stakeholder understanding and technical implementation.

**Domain Model** At the highest level of abstraction, class diagrams are used to model concepts from the problem domain. These diagrams represent real-world entities and their relationships, often using language familiar to stakeholders. The focus here is on understanding and communicating the essential structure of the domain.

**Logical Model** As the design progresses, the domain model is refined into a logical model. This level introduces more technical detail, such as class responsibilities, access modifiers, and associations with navigability or multiplicity. It provides a blueprint for the software's internal design while remaining independent of specific implementation technologies.

**Physical Model** Eventually, the logical model is translated into a physical model, where class diagrams may reflect actual code structures, database schemas, or deployment components. At this stage, technical constraints and implementation details (such as data types or platform-specific features) are included.

Class diagrams also share conceptual similarities with Entity-Relationship (ER) diagrams, which are commonly used in database design. In fact, UML class diagrams can serve as a richer alternative to ER diagrams, especially when modeling object-oriented systems:

- Entities in ER diagrams correspond to classes in UML
- Attributes are represented similarly
- Relationships in ER diagrams map to associations, aggregations, or compositions in class diagrams, often with multiplicity annotations

- UML also supports advanced features such as inheritance and method specification, which are typically absent in ER diagrams

### 2.3.2 Components of Class Diagrams

A class diagram consists of multiple components. In broad terms, those components are classes, their attributes and operations, and the relationships between different classes.

Classes and objects are closely related. A class is a blueprint for creating objects - it describes what attributes and operations an object is going to have and how it is going to interact with other entities.

An object is an instance of a class. For example, a class Table is an abstract idea that is used for creating real tables. A table in the local library is an object - a specific instance of a table that follows the specifications described in the class Table.

#### Attributes

Attributes describe the properties of a class. A class defines the list of attributes along with their data types and visibility. The class may also define an attribute's multiplicity - the number of values that are to be assigned to it. The default amount is 1. In the case of an object, the attributes are assigned specific values.

#### Operations

Operations are activities that instances of a class can perform. They include things like performing calculations or altering a value. A class describes the visibility, input parameters, return type, and the name of the operation.

#### Data types

Data types define what kind of information can be stored in a variable (or returned by an operation). Examples are integer, character, string, or even an instance of another class.

#### Visibility

Visibility sets which entities are allowed to view and interact with an attribute or an operation. There are four types of visibility - public, protected, default and private.

#### Modifiers<sup>3</sup>

Modifiers in UML class diagrams provide additional information about the behavior and constraints of classes, attributes, and operations. Common modifiers include abstract, which indicates that a class or method cannot be instantiated or directly invoked and must be extended or overridden; final, which marks a class as not extendable or

a method as not overridable; and static, which signifies that an attribute or operation belongs to the class itself rather than to any instance of it.

UML also supports the interface modifier to represent interfaces—types that declare operations without implementing them, which classes can then realize. These modifiers help clarify the intended usage, inheritance, and design principles of class components in object-oriented modeling.

### Example of a class<sup>3</sup>

The following figure displays an example of a class from a class diagram. It models a simple User entity that has credentials and a birthdate, allowing the system to update the password and email. It demonstrates the key features of a class, such as attributes, operations, types, operation parameters or access modifiers.

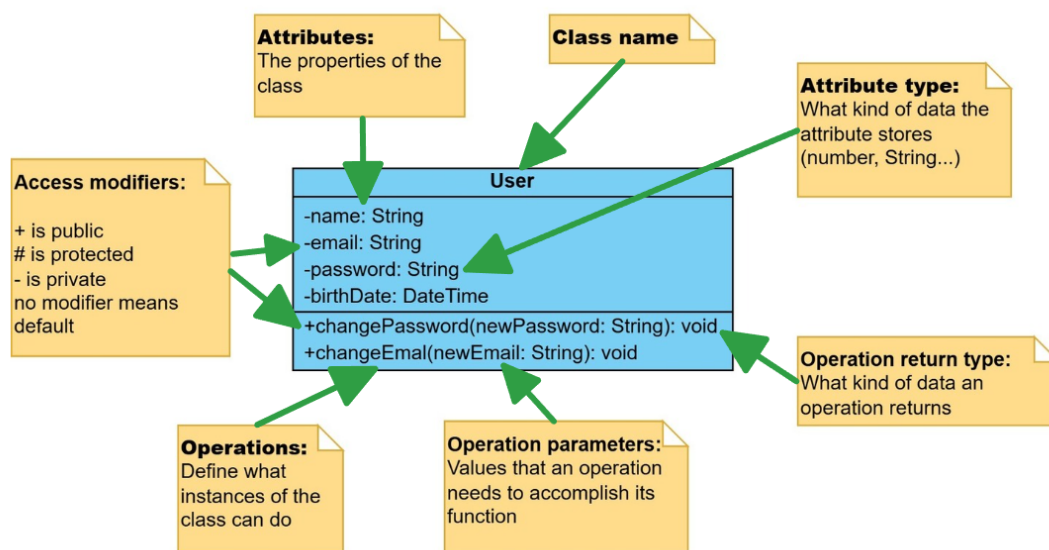


Figure 2.2: An example of a class

Made with: "Visual Paradigm," *Visual Paradigm Online*, <https://online.visual-paradigm.com/>, accessed May 23, 2025.

## 2.4 Relationships<sup>3</sup>

The class diagram defines several types of relationships between different classes, each representing a distinct kind of interaction or connection within a system.

### Association

Association represents a general connection between two or more classes. It indicates that objects of one class are connected to objects of another. This is the most basic relationship and can be bidirectional or unidirectional.

### Aggregation

Aggregation is a specialized form of association that represents a whole-part relationship between the aggregate (whole) and its parts. Unlike composition, aggregation implies a weaker relationship where the parts can exist independently of the whole.

### Composition

Composition is a stronger form of aggregation. It also represents a whole-part relationship, but with the key difference that the part cannot exist independently of the whole. If the whole is destroyed, its parts are typically destroyed as well.

### Inheritance

Inheritance (also known as generalization) represents an "is-a" relationship between a more general superclass and a more specific subclass. The subclass inherits the attributes and behaviors of the superclass, allowing for code reuse and polymorphism.

### Dependency

Dependency indicates a relationship where one class depends on another because it uses it temporarily, such as by calling its methods or using its data. It is typically a weaker relationship and often represents a "uses-a" connection.

### Realization

Realization is a relationship between an interface and a class that implements that interface. It shows that the class provides concrete implementations for the methods defined in the interface.

### Cardinality

Cardinality tells how many instances of a class are to be associated with an instance of another class. The most common cardinalities are one-to-one, one-to-many or many-to-many.

## 2.5 Example of a Class Diagram<sup>3</sup>

The following figure displays an example class diagram. It represents a simplified model of an e-shop, where customers place orders and admins manage product catalogs. Orders contain multiple items and can be paid using various payment methods.

It demonstrates many of the essential concepts of a class diagram, mainly inheritance, aggregation, composition, association, multiplicity, realization, and classes with their attributes and operations.

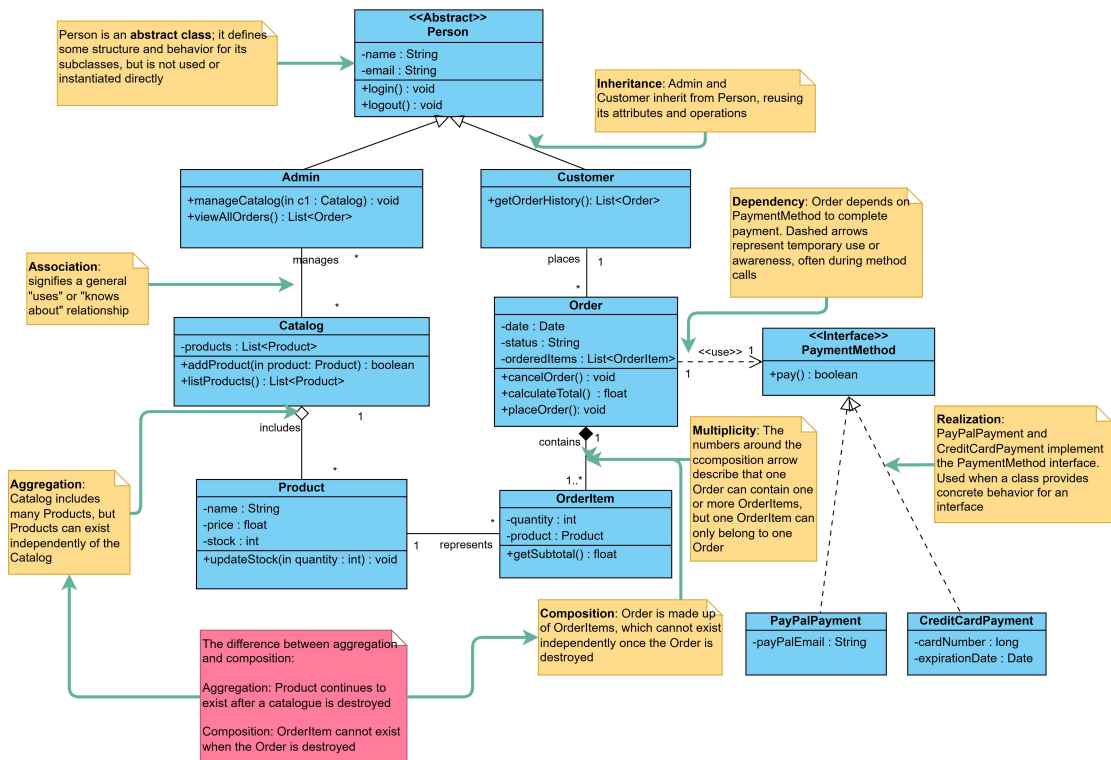


Figure 2.3: An example of a class diagram.

Made with: Visual Paradigm, *Visual Paradigm Online*, <https://online.visual-paradigm.com/>, accessed May 23, 2025.





## Chapter 3

# E-learning and the UML Class Diagram

### 3.1 Introduction to E-learning<sup>1</sup>

E-learning, or electronic learning, is a modern approach to acquiring knowledge and skills through digital technologies. It is a rapidly growing field that offers a wide range of possibilities for making education more effective, accessible, and flexible. With the help of the internet, multimedia tools, and interactive systems, learners can study anytime and from anywhere, overcoming many of the limitations of traditional classroom education.

E-learning is used in various forms—ranging from simple online courses and video lectures to complex platforms that offer feedback, progress tracking, and personalized learning experiences. It often complements face-to-face teaching in a blended learning model, which combines the strengths of both in-person and digital instruction.

Among the key benefits of e-learning are improved accessibility, self-paced learning, and the potential to tailor educational content to individual needs. However, it also faces challenges such as reduced social interaction, learner isolation, and varying levels of digital literacy. Despite these issues, the field continues to evolve, with current trends focusing on practices like mobile learning, gamification, microlearning, and adaptive learning technologies.

As digital tools become more widespread and learning needs become increasingly diverse, e-learning is likely to play an even more significant role in the future of education.

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<sup>1</sup>This section was made with the use of Gen AI

## 3.2 Comparison of E-learning and traditional learning

Traditional learning refers to the model where a teacher or a tutor presents information and exercises to a group of people in a classroom. This model is typical of educational institutions such as universities. This approach has been utilized for a very long time and it is the standard way of teaching.

E-learning offers a number of improvements over the traditional model. Unlike traditional learning, online courses can be taken from anywhere and at any time, removing much of the geographic and time constraints for the learners. It is also significantly cheaper, since it removes the need for a teacher or a learning space. With interactive exercises and materials, e-learning can increase student engagement and interest in the topic.

On the other hand, it also presents a set of challenges that make traditional learning still the preferable choice in most cases. The most obvious drawback is that the self-directed nature of e-learning requires a level of discipline from the learners which many lack. Other problems include a lack of social interactions with other students, a lack of personalized feedback from a live instructor and the fact that some people do not have access to the technology necessary to utilize e-learning materials. For some people, digital literacy may also be a problem stopping them from using e-learning resources.

In a study performed by Ramle et al.[4] a UML course was introduced. Students that took the course were tasked by filling out a questionnaire about their experience. The reception was mostly positive, with students making claims such as *“makes me understand more”* or *“makes you get better at UML.”*

Jurgelaitis et al.[5] found that the introduction of gamified mechanics to a university UML course increased both students’ motivation and their average scores. This confirms the general expectations from other works that have not yet collected the necessary data for evaluation.

Lastly, Soler et al.[6] proposed a tool for automated correction of class diagram exercises. The researchers conducted an experiment where students were divided into 2 equally sized groups: one solved a set of exercises using the tool, and the other solved them manually. The number of students who solved all of the exercises correctly was higher in the group that used the tool, and this group also had better average score in a subsequent exam.

These studies clearly demonstrate that tools can be developed to increase not only the students’ knowledge, but also their engagement and overall satisfaction with their educational process.

## 3.3 Techniques in E-learning<sup>1</sup>

### 3.3.1 E-learning techniques

There are many techniques and practices that can be used to enhance the learning process when it comes to e-learning materials. Some of them are aimed at increasing the learner's motivation and interest (gamification), some directly help improve and better organize the study materials (microlearning, adaptive learning) and some of them are for convenience (self-directed learning). All of them have their use and when applied correctly, they can have a positive impact on the overall learning experience of the learners.

#### **Gamification**

Gamification involves integrating game-like elements into the learning process—such as points, levels, badges, leaderboards, and challenges—to increase engagement and motivation. By leveraging learners' natural desire for competition and achievement, gamification can make education more interactive and enjoyable, especially in areas that may otherwise seem abstract or technical.

#### **Microlearning**

Microlearning is based on delivering content in small, easily digestible chunks. Each module typically focuses on a single concept or skill and can be completed in a short amount of time. This approach is well-suited to modern learners with limited attention spans or tight schedules, and it allows for quick revisions and focused repetition.

#### **Self-directed Learning**

Self-directed learning empowers students to take control of their learning paths by setting their own goals, choosing resources, and pacing their progress. While this method requires a high level of motivation and discipline, it fosters independence and lifelong learning skills. E-learning platforms often support SDL through customizable paths and open access to resources.

#### **Blended Learning**

Blended learning combines traditional face-to-face instruction with digital learning methods. This hybrid model takes advantage of the strengths of both approaches—personal interaction and immediate feedback from in-person classes, along with the flexibility and scalability of online learning.

#### **Adaptive Learning**

Adaptive learning systems use algorithms and data analytics to tailor content and as-

assessments to the learner's individual performance and needs. By identifying strengths, weaknesses, and learning patterns, these systems provide a personalized experience that can improve outcomes and maintain engagement.

### **Video-based Learning**

Using videos as the primary mode of instruction is a popular and effective technique in e-learning. Videos can explain complex concepts visually, combine audio and text for better retention, and allow students to pause, replay, or speed up content based on their preferences.

### **Social Learning**

Social learning is based on the idea that people learn effectively through observation, imitation, and interaction with others. In e-learning environments, this is facilitated through social features such as discussion forums, comment sections, peer reviews, live group sessions, and integration with social media. These tools allow learners to share knowledge, ask questions, provide feedback, and learn from each other's experiences. Social learning fosters a sense of community and can increase motivation by creating a more collaborative and connected learning environment.

### **3.3.2 Impacts of various e-learning techniques on students**

There are many ways these techniques can be implemented, and they do not have the same effect on the learners. Using too many is likely to overwhelm the students and worsen their learning experience, so it is important to choose only a limited number of them.

There is a lot of research done that confirms that e-learning can indeed increase the students' engagement, motivation and learning effectiveness . There is much less research concerning comparison of the individual e-learning techniques in these regards.

references

Different aspects of e-learning can be examined in this way. A study [7] suggests the type of feedback delivered after each exercise can significantly impact the students' motivation. Apart from being immediate, feedback can take various forms. The study reveals the most potent form is providing not only a confirmation of the result (correct/incorrect), but also explaining why the answer is correct or incorrect, increasing both the students' motivation and objective performance in tests more significantly than other feedback approaches.

Another study [8] compares the effects of gamification and personalization in e-learning environments. By gamification, it means immersion practices such as avatars, profiles and storytelling/role-play, social features (teams, cooperation, competition) or the

more achievement based techniques such as leaderboards, challenges, points and rankings.

Personalization means tailoring content to the students' individual needs by providing self-paced environment, personalized feedback from instructors or personalized content selection or even adaptive learning paths. This can be described as adaptive learning. The recommendations can be based on user data like their major or fields of interest, their behavior or in the types of mistakes they make, among other things.

The key findings of the study are that while personalization drives cognitive learning engagement, gamification promotes emotional learning engagement. Though not surprising, the implications are personalization improves learning efficiency, while gamification can be applied to keep the students engaged and persist in the process. Gamification is then more of a psychological than objective factor.

Another important finding is that different studied gamification elements have varying effect. The social elements (collaboration, competition) and immersion elements (avatars, storytelling) were of higher importance than achievement elements (experience points, badges). It does state the achievement elements are useful nonetheless.

Though the reception of gamification is positive in almost all research papers we reviewed, some works suggest it is not so efficient in all instances[9]. However, the study aggregates many research papers across various fields, target audiences and inconsistent gamification implementations. Despite potential issues with the study, it shows that gamification is not automatically an improvement to study materials and its effectiveness depends on context and its implementation.

This point is further supported by Luo et al.[10]. It sums up other research in which gamification was not implemented successfully. The failures are linked to shift from intrinsic motivation (interest in learning) towards earning points and rank. Other potential issue was poor implementation of the gamification elements, making the students more confused than motivated.

Luo et al.[10] mentions that among the most significant barriers to gamification implementation were "Inappropriate or superficial use of gamification techniques" and "Poor integration of game features and pedagogical content".

### 3.4 Applying E-learning to UML Class Diagrams

Since the experience of studying computer science contains a large amount of self-study in any case, there are many online resources and courses in the field. Some of them utilize the techniques mentioned in the previous section. We investigated how they

This is a fishy study. If i am to disregard the findings in one case, I cannot accept them in other instances. Ew, delete this I guess

warnings about competitive activities?

were used in more detail to determine whether such practices could be relevant for our intended course.

### 3.4.1 Gamification

Gamification is the most commonly mentioned practice when it comes to research of software modeling e-learning, and there are numerous attempts at applying it in the context of teaching the UML class diagram. The goal is to increase the students' interest and motivation in the given tasks by exploiting their natural drives.

In *Actionable Gamification: Beyond Points, Badges, and Leaderboards* [11], Yu-kai Chou introduces the Octalysis Framework, a comprehensive model for analyzing and designing engaging experiences through gamification. The framework is built around eight core drives that motivate human behavior, such as meaning, accomplishment, empowerment, ownership, social influence, scarcity, unpredictability, and avoidance. Unlike traditional gamification approaches that focus mainly on superficial rewards, Octalysis emphasizes the psychological factors behind user motivation, enabling the creation of more meaningful and effective game-like experiences in both digital and non-digital contexts.<sup>1</sup>

The practice of gamification involves several techniques. Cagnazzo et al.[12] mark a student's accomplishment by levels - the student starts at level one and is awarded experience points for completing exercises. The amount of experience points gained is affected by the difficulty of the given exercise as well as the number of mistakes made by the student, providing additional motivation to reduce the mistakes for the student. After gaining enough experience points, the student's level increases. The study proposes even the introduction of quests to further engage the learners.

The same study also implements a customizable avatar that represents the student in the application. They can alter the avatar's appearance by changing its clothes, hairstyle and other visual features. Completing exercises unlocks new styling options, again increasing the student's motivation. Lastly, the tool immediately displays feedback to the student by highlighting mistakes in various colors and by changing the student's avatar's facial expression based on the student's performance.

In another work, Jurgelaitis et al.[5] implement, in addition to previously mentioned levels and experience points, badges that are awarded for accomplishing specific goals and coins that can be exchanged for certain items. The application displays immediate feedback after completing tasks and visualizes the student's progress. Leaderboards with students ranked by their experience points are also included, invoking the students' natural competitiveness.

There are other options to incorporate the concepts of gamification into the teaching of

modeling. Besides some of the already described techniques, Bucchiarone et al.[13] connect modeling with the hangman game - each mistake draws another part of the hangman, until either the exercise is completed or the hangman picture is finished (which means the player has lost).

### 3.4.2 Microlearning

Microlearning is the practice of breaking the study material down into small, bite-sized pieces so as to avoid overwhelming the learners. Ramle et al.[4] mention that the Learn section of their application divides the study materials into smaller sections. The general reception of the application was positive and students claimed that it helped them understand the concepts better. That suggests that applying microlearning to the topic of UML may improve the students' learning experience and possibly their resulting knowledge as well.

However, it should be noted that, even though microlearning improves motivation and knowledge retention, it is more suitable for early stages of learning when basic concepts are explained. It might not be the ideal approach when tackling more complex concepts or when constructing the bigger picture out of the smaller lessons<sup>2</sup>. Therefore, microlearning should not be applied to every concept, but rather to the earlier lessons only.

### 3.4.3 Social learning

Ramollari et al.[14] designed a system in which students were encouraged to discuss and share their solutions to modeling exercises. Besides discussions and comments, the students could attach their solutions to other students' work as hints, alternatives or suggestions for improvement. The efforts were evaluated positively.

## 3.5 Learning Management Systems (LMS)<sup>1</sup>

LMS have become an increasingly popular choice for educational institutions and individual educators seeking flexibility, transparency, and cost-effective solutions. These platforms offer a wide range of features to support online learning, from course management and content delivery to assessment and communication tools.

In this section, we briefly examine several prominent open-source LMS platforms, evaluating their general characteristics, strengths, and limitations. Each system offers a

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<sup>2</sup>Christopher Pappas. "Microlearning in Online Training: 5 Advantages and 3 Disadvantages." *eLearning Industry*, March 15, 2016. Available at: <https://elearningindustry.com/microlearning-in-online-training-5-advantages-and-3-disadvantages>. Accessed May 15, 2025.

unique set of features. We will perform a comparison based on our assumed requirements that will be described later in section.

### **Moodle<sup>3</sup>**

Moodle is a widely-used and feature-rich open-source learning management system that supports a broad range of teaching styles and learning needs. It offers powerful tools for content delivery, assessment, tracking learner progress, and supporting collaborative learning through forums and wikis. Thanks to its extensive plugin ecosystem, Moodle supports gamification, microlearning, adaptive learning, and integration with external tools. While highly customizable, its interface and complex configuration may feel overwhelming to users seeking simplicity.

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<sup>3</sup>"Moodle," *Moodle Project*, <https://moodle.org/>, accessed May 25, 2025.



### **Open edX<sup>4</sup>**

Open edX is a scalable and modular LMS originally developed by Harvard and MIT. It is especially strong in supporting self-paced and adaptive learning through structured content, video lectures, quizzes, and analytics tools. Its powerful authoring tool, Studio, allows course creators to design interactive and engaging learning experiences. While Open edX is ideal for research-driven or large-scale deployments, it has a steeper learning curve and requires more technical resources to deploy and maintain.

### **Chamilo<sup>5</sup>**

Chamilo is an open-source LMS focused on ease of use and fast deployment, making it suitable for institutions prioritizing simplicity over advanced customization. It includes built-in tools for course management, assessments, certification, and collaboration. Chamilo enables tracking student progress and provides basic support for gamification and interactive learning. Its clean interface and low technical demands make it a solid choice for smaller teams or pilot e-learning projects, though it offers fewer integrations than larger platforms.

### **ILIAS<sup>6</sup>**

ILIAS is a mature, flexible LMS that supports complex learning scenarios, including competency-based education, learning paths, and detailed tracking of user progress. Originally developed for academic institutions and public organizations, it emphasizes structure, reusability of content, and rich assessment tools. ILIAS also includes collaborative tools such as blogs, forums, and group workspaces. While powerful, its interface can appear outdated and may require configuration to meet modern usability expectations.

### **Canvas LMS<sup>7</sup>**

Canvas LMS offers a modern, user-friendly experience with a focus on intuitive design and ease of use. Developed by Instructure, its open-source version includes core features such as course creation, grading, analytics, and communication tools. Canvas supports REST APIs, making it easy to extend or integrate with external tools like custom diagram editors or assessment modules. Though some advanced analytics and AI features are exclusive to the commercial version, the open-source release remains highly capable and suitable for most academic settings.

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<sup>4</sup>"Open edX," *Open edX Project*, <https://openedx.org/>, accessed May 25, 2025.

<sup>5</sup>"Chamilo," *Chamilo Association*, <https://chamilo.org/en/>, accessed May 25, 2025.

<sup>6</sup>"ILIAS," *ILIAS Open Source e-Learning*, <https://www.ilias.de/en/>, accessed May 25, 2025.

<sup>7</sup>"Canvas LMS," *Instructure*, <https://www.instructure.com/canvas>, accessed May 25, 2025.

**Overall comparison**

| Feature                                  | Moodle                | Open edX            | Canvas LMS            | ILIAS                 | Chamilo       |
|------------------------------------------|-----------------------|---------------------|-----------------------|-----------------------|---------------|
| <b>Simplicity (UI/UX)</b>                | Moderate              | Moderate            | High                  | Low                   | High          |
| <b>Progress Monitoring</b>               | Yes                   | Yes                 | Yes                   | Yes                   | Yes           |
| <b>Customization</b>                     | Very High             | High                | Moderate              | Moderate              | High          |
| <b>Ease of Content Extension</b>         | High                  | High                | Moderate              | Moderate              | High          |
| <b>Gamification</b>                      | Yes (plugins)         | Limited             | Yes (plugins)         | Yes                   | Yes           |
| <b>Microlearning</b>                     | Yes                   | Yes                 | Moderate              | Yes                   | Yes           |
| <b>Adaptive Learning</b>                 | Yes (plugins)         | Yes                 | Limited               | Limited               | Moderate      |
| <b>Community and Plugin Availability</b> | Very Large            | Large               | Moderate              | Moderate              | Growing       |
| <b>Feedback Channels</b>                 | Yes                   | Yes                 | Yes                   | Yes                   | Yes           |
| <b>Ideal Use Case</b>                    | Schools, Universities | MOOCs, Corporations | Schools, Universities | Corporate, Government | Schools, NGOs |

Table 3.1: Comparison of selected open-source Learning Management Systems

### 3.6 Common issues in Learning UML

There are several common challenges students face when learning UML. Some of them are not encountered by students exclusively but also by graduates and practitioners.

Fresh computer science graduates are expected to possess at least a basic understanding of the use of UML in software development, but that is not often the case [15]. In addition to that, even more experienced practitioners or programmers commonly claim to know UML, but later prove to only have a very limited or even wrong understanding of the concepts<sup>8</sup>.

This gap arises because UML is widely recognized as complex and difficult for novices to grasp. Studies have consistently shown that students struggle with selecting the appropriate level of abstraction, applying correct syntax and semantics, and translating mental models into UML diagrams. Moreover, the same recurring mistakes continue

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<sup>8</sup>Donald Bell, "An Introduction to the Unified Modeling Language," *IBM Developer*, <https://developer.ibm.com/articles/an-introduction-to-uml/>, accessed February 23, 2025.

to be observed over the years, indicating persistent conceptual misunderstandings and difficulties in mastering the modeling process itself [16]<sup>1</sup>

There are multiple research papers describing the process of identifying errors made in class diagrams. The studies [17], [18], [16] made catalogues of such mistakes, allowing easier identification of errors and potentially using them as educational materials. Chren et al.[18] revealed that the most common mistakes were related to associations, lack of domain understanding, applying inappropriate levels of detail, placing operations into wrong classes or incorrect use of inheritance.

Reuter et al.[16] provided a comprehensive taxonomy of problems encountered by students, including confusion between class diagrams and other UML diagram types, difficulties in determining appropriate attributes, operations, or classes, uncertainty in modeling associations and their properties, and problems distinguishing between relationships such as inheritance, realization, aggregation, and composition<sup>1</sup>.

These studies highlight the need to improve the way UML is taught in most universities, as the students' knowledge often appears insufficient. Areas for improvement include providing more practical exercises, immediate feedback, increasing students' engagement, or connecting modeling with coding.



## Chapter 4

# Analysis summary

### 4.1 Related work

There are multiple studies that tried to create e-learning modules or tools aimed at enhancing the teaching of the UML class diagram. They applied various approaches and ideas to the task.

#### **Automated diagram assessment tools**

Much of the other researchers' efforts were aimed at creating modeling tools that would enable students to draw their own class diagrams from scratch and then automatically evaluate their submissions. The goal is to provide fast and personalized feedback to the students and reduce the teachers' workload.

In these studies, the requirements for those tools were fairly similar; the drawing tools should be simple and intuitive and should be able to export the created diagrams in a format compatible with the evaluation algorithm (XMI in most cases). Then the diagram would compare the student's solution with a premade example solution and list errors or recommendations for improvement for the student.

One of the main issues with automated class diagram assessment is the fact that a modeling task is very likely going to have multiple correct solutions. One way of dealing with this issue is to store a number of possible solutions for each given problem and use the solution that is most similar to the student's current submission [6]. However, it is also possible to only use one sample solution. Doing so presents the risk of providing misleading feedback, missing errors or marking correct options as errors. In practice, these issues do not seem to be significant [19]. A less common approach to diagram quality assessment was using machine learning which achieved promising results, though only provided grading of the diagrams without specific feedback [20].

Modi et al.[21] described a tool for drawing and automatically assessing submitted diagrams. Unlike most other tools of the kind, which only support one type of diagram, the proposed tool supports a number of various UML diagrams, including the class diagram. The study also includes a brief comparison of similar tools and concludes that such tools can decrease the time and effort needed from teachers when evaluating students' diagrams.

Schramm et al.[19] also introduced an automated tool for correcting students' diagrams by comparing them to an example solution. The results of the study did not suggest that using an automated system for correcting diagrams yields better results than relying on feedback from a teacher. However, they demonstrate that such a system can save the teachers' time and resources without compromising the students' learning process.

### **E-learning applied to the UML Class Diagram**

There were other studies aimed at teaching the Unified Modeling Language class diagram using other techniques than an automated tool for correction of the diagrams. They created platforms with various forms of quizzes and exercises and some of them applied e-learning practices such as gamification or microlearning.

In a study performed by Ramle et al.[4], a simple e-learning application for teaching several types of diagrams was designed and tested on a small sample of students. The section dedicated to explaining the concepts contains small-sized bits of information as to avoid overwhelming the students (an example of the use of microlearning). The application was generally well received by the students and, according to a survey filled by the participants, also helped them understand UML better.

Cagnazzo et al.[12] proposes using gamification mechanics to enhance the modeling classes. It introduces a web-based tool that allows the students to draw diagrams and have them automatically evaluated for mistakes. The gamified practices include levels, where a student's accomplishment is determined by a numeric value. The level can be increased by solving exercises. Another mechanics implemented in the tool are offering the students the option to customize their avatars (a picture that represents them in the application) and visually enhanced feedback that highlights their mistakes.

There are more proposed enhancement for the teaching of UML. Almadi et al.[22] propose creating a recommender system to suggest the most relevant modeling exercises to students. Lethbridge et al.[23] describe how using tools that convert diagrams to code and vice versa in classrooms helps students understand the abstract concepts of modeling. Shmallo et al.[17] arranged an activity during which students were to find mistakes in each other's diagrams. Such an activity significantly improved both their modeling skills and motivation.

Despite often being mentioned as prominent gamification elements, the concepts of storytelling and inducing competition among participants in context of software modeling e-learning materials are not discussed in the reviewed literature.

| Study                   | Notable Features                                                                                               | Key Findings                                                                                                                                                                     |
|-------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ramle et al. [4]        | A course on UML, microlearning, self-directed learning                                                         | Well structured, interactive exercises and self-paced approach in the course were well received by students and improved their understanding of UML                              |
| Jurgelaitis et al. [5]  | A gamified online course used in a university software modeling class                                          | Gamification increased both students' motivation and their average scores                                                                                                        |
| Soler et al. [6]        | An automated diagram assessment tool used in a classroom                                                       | The group who used the tool scored higher in an exam than the group who used any other resource of choice                                                                        |
| Cagnazzo et al. [12]    | An automated diagram assessment tool, learning platform with gamified elements                                 | Instant feedback on modeling exercises and gamified elements improved both student engagement and modeling skills                                                                |
| Bucchiarone et al. [13] | Online learning tool applying gamified mechanics                                                               | Using a gamified learning tool improved motivation and results of the students. They also found using the tool enjoyable                                                         |
| Ramollari et al. [14]   | Collaborative learning environment – students discussed, evaluated and helped each other in modeling exercises | The extensive collaboration helped the students comprehend the taught materials, found it to be an effective learning method and the feedback on the system was largely positive |
| Shmallo et al. [17]     | Identification of mistakes in diagrams as a collaborative exercise (not an e-learning tool, however)           | Searching for mistakes in each other's diagrams and discussing them both engaged the students and improved their modeling skills                                                 |
| Schramm et al. [19]     | An automated diagram assessment tool                                                                           | The proposed automated diagram assessment tool saved teachers' time and energy without reducing the quality of provided education                                                |
| Almadi et al. [22]      | Recommender system to propose exercises to students                                                            | Personalized exercise recommendations help students better understand UML and promote self-directed learning                                                                     |
| Lethbridge et al. [23]  | Modeling tool that converts diagrams to code and vice versa                                                    | Students understand UML better when they see code corresponding to the diagrams                                                                                                  |

Table 4.1: Summary of notable e-learning tools and methods for teaching UML and software modeling



## 4.2 Summary

UML is a language used for modeling systems from various perspectives. The class diagram is the most prominent of the UML diagrams. It is used in various stages of system development as well as for documentation purposes.

For these reasons UML is a common part of the curriculum at universities related to computer science. Despite its inclusion, it is often observed that students as well as graduates and even professionals struggle with UML. To solve this problem, it is a worthwhile effort to improve the education of software modeling in universities.

Attempts have been made to improve the situation with the use of e-learning - the use of digital technologies to deliver educational content. E-learning can provide numerous benefits such as increased student engagement, instant personalized feedback or saving the time and attention of teachers. It is also possible to use e-learning as an addition to traditional classrooms rather than their replacement - such practice is called blended learning.

There are numerous commonly used e-learning practices with various intended benefits. The ones most suitable for educational content related to software modeling are gamification, microlearning, blended learning and social learning.

We analyzed several studies that implemented these techniques in UML learning tools. The most popular technique was gamification, where the tools support functions resembling games. Examples are giving experience points or some sort of currency for completing exercises and allowing students to create customizable avatars to represent them in the learning environment.

Most of the studies were aimed at creating tools to enhance the in-person classes at universities, not to replace them. Some of them presented courses with study materials, practical examples and tests; some introduced new collaborative learning activities, and some studies proposed tools for automated diagram assessment.

The results of these studies were predominantly positive, showing improvements both in students' scores and their engagement with the course topics. Another benefit of the described tools is that they free up the educators' time and energy that they would otherwise spend reexplaining concepts and correcting exercises.

These findings clearly document the potential of introducing e-learning tools to enhance the teaching of UML in university courses related to software modeling.

Storytelling, role-play and competition are commonly mentioned as prominent e-learning elements. Despite that, the reviewed literature regarding application of e-learning to software modeling did not discuss them. That creates a gap in research that we seek to

fill out.

Based on the analysis, we decided to create an online course aimed at the UML class diagram. It will implement some of the e-learning techniques, most notably gamification and microlearning, and will test it out on students of a software modeling subject at FIIT STU. We will focus on the storytelling and role-playing aspects of gamification, using them to create 2 different designs of the course: a competitive one, and a playful one. Both objective test results and students' subjective opinions will be evaluated and used to compare the two designs.

Depending on the reception of the project by students, further research can be performed to extend the scope of the course. It may include other types of diagrams or software modeling materials, other areas of software engineering, or extend the use of the e-learning practices.

place for hypothesis if we need any





## Chapter 5

# Objectives and methodology

### 5.1 Objectives

The goal of this study is to create an online course for teaching the UML class diagram and analyze various approaches to doing so. The target audience are computer science students taking a software modeling course at FIIT STU.

It is a project that aims to compare 2 of the most prominent approaches to enhancing engagement and information retention in e-learning - competition and play - and applying them to teaching software modeling. We want this study to be a starting point for further educational content at FIIT STU should this project prove successful. In such a case, other researchers could extend the course by adding additional features and content, including other types of diagrams in it, or by adjusting the materials to try out different e-learning approaches to delivering content and driving engagement. On top of that, e-learning materials in other areas might be developed using findings in this project.

We present the following research questions based on our initial analysis and objectives:

1. Does competitive course design contribute to students' error reduction and higher overall performance more than playful course design?
2. Does playful design lead to higher behavioral engagement?
3. How do students subjectively evaluate the playful and competitive approaches to delivering study materials?
4. Is it worthwhile to create additional e-learning content at FIIT STU?

## 5.2 Methodology

The courses will be implemented in an established e-learning platform. They will contain theoretical materials, interactive exercises, self-assessment components, as well as e-learning practices based on our analysis. Both courses will contain the same exercises and materials, but they will differ in tone and use of e-learning practices. That will allow for separate analysis of the educational approaches.

The courses will be taken by second-year students of computer science who will be presented with 2 tests and 2 questionnaires - one of each at the start, and one of each at the end of the courses. They will be used to measure the students' interest and level of knowledge before and after taking the course. Each student will only take 1 of the courses.

The tests will be made up of theoretical questions and practical exercises, while the questionnaires will be aimed at the students' subjective reception and interest. The most important questions will target the students' interest in such materials and whether they think it is an improvement over the standard way of teaching software modeling.

The students will be divided into two groups: one for each of the courses. Both groups will be given the same tests and questionnaires to determine their skill level and subjective opinions. We will gather results, analyze them and draw conclusions based on them.







# Chapter 6

## Design

### 6.1 Solution description

When designing our intended class diagram courses, we will take into account a number of requirements. Firstly, we have the platform or LMS on which to implement the course, then there are e-learning techniques to use, specific types of exercises and other educational content we want to apply, and lastly we need to propose an effective and objective evaluation method to determine the potential of the project for further research.

Additional constraint that we need to account for is the scope of the course. Even though an ideal UML course might guide the user through building an entire system, this course is complementary and the software modeling class's curriculum does not make space for it. Participation in this course is voluntary. For these reasons, we decided to restrict the course's scope to only one diagram type to test out the concept, and to fit all the content into one hour.

The UML class diagram was preferred over other diagram types, because it is one of the most prominent UML diagram types - likely the most widely used one - and it is intuitive to understand. It directly links to coding which students tend to comprehend better than abstract diagrams [23].

#### 6.1.1 E-learning platform

We want to create the courses on a preexisting LMS. Doing so will significantly reduce development time, allowing us to focus on the educational content.

The platform should satisfy the following conditions:

- easy to use and learn

- record and store students' scores and progress for result comparison and material recommendations
- option to propose changes and improvements, report mistakes and bugs
- easy to include new features and content
- free to use or open source
- support for e-learning techniques described in the next sections
- support for interactive exercises

### 6.1.2 Content Structure

This section is aimed at describing what specific types of content the course should contain. We will divide it into four sections - an introductory test to assess the starting point of the students, a theoretical section dedicated to explaining concepts, one for practical exercises, and one for final skill assessment after taking the course. The final test will be accompanied by a questionnaire.

1. Introductory assessment test
2. Theory
3. Practical exercises
4. Final assessment test, questionnaire

### Activity diagram - student

The activity diagram describes a student's interaction with the system. The student will log in to the system and view a course. They will complete the introductory test. Then they will proceed to the theoretical section with micro-exercises in between modules and the practical section containing more exercises. Content will be recommended based on how successful their responses were across various topics. The student can come back to the theoretical section and the exercises and redo them until they are satisfied with their results. Lastly, the student will enter the evaluation section, where they will complete a test (which will not give them instant feedback this time), and fill in a feedback questionnaire.

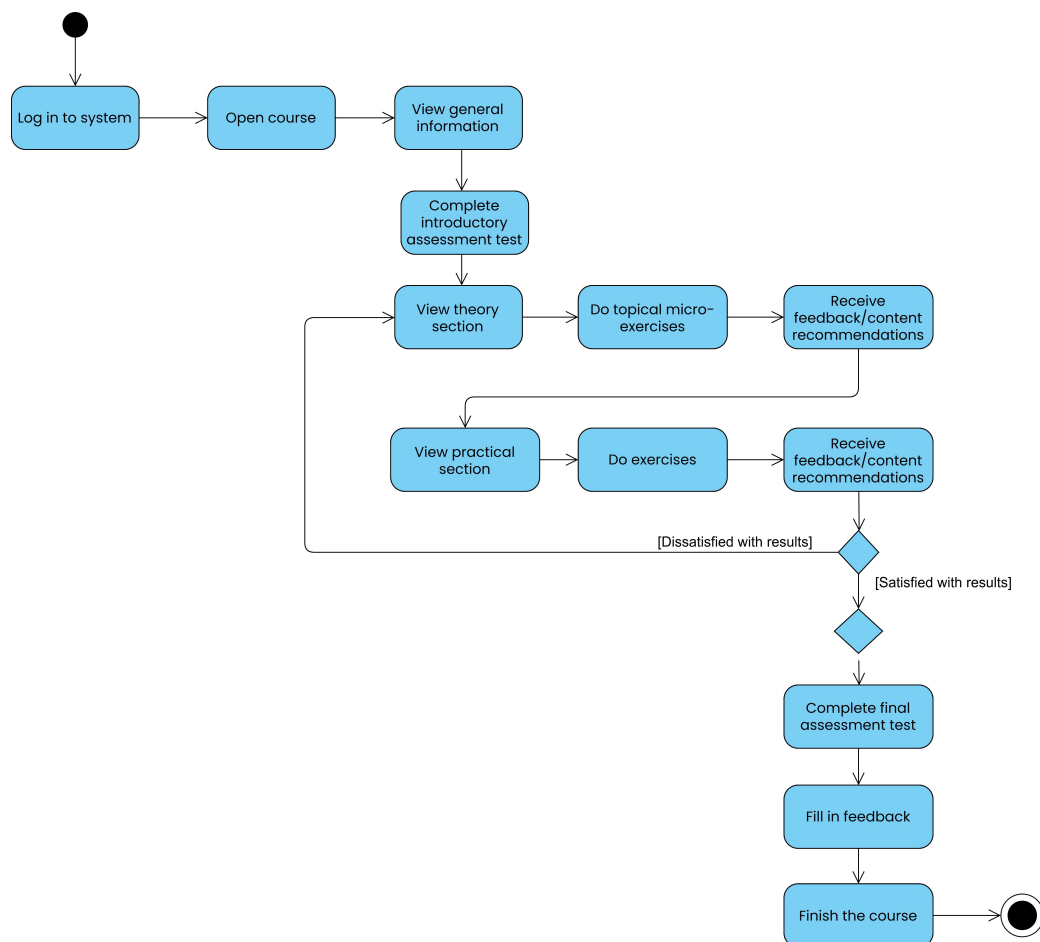


Figure 6.1: Activity diagram from a student's perspective.

Made with: Visual Paradigm, *Visual Paradigm Online*, <https://online.visual-paradigm.com/>, accessed Dec 9, 2025.

### 6.1.3 Content - Theory

The theoretical section will contain a set of chapters each dedicated to a single concept or component of the class diagram. The concepts will be added gradually, simulating the process of building a diagram from scratch, while describing modeling decisions at each step. There will be micro-exercises placed between theoretical sections.

Since students understand UML better when they see corresponding code [23], there will also be a snippet of code available for diagram demonstrations.

### 6.1.4 Content - Exercises

A limitation of the exercise design is the absence of a UML diagram editor. We do not have the capacity to employ teachers to manually assess modeling exercises. At the same time, designing a comprehensive automated diagram assessment tool is a non-trivial problem and is beyond the scope of this project.

For these reasons, we will replace direct modeling tasks with a set of exercises implemented using more general, readily available formats. The exercise design is inspired by findings from our analysis, namely:

- Searching for mistakes in others' diagrams can enhance the students' modeling skills [17]
- Students develop a better understanding of UML when they see diagrams translated to code [23]
- Exercises work best when feedback is immediate and explains why the answer is correct or incorrect [7]

They will be of the following types:

- Fill in missing parts of a diagram (relationships, multiplicity...). This is the most important exercise type, since it directly simulates modeling
- Assign the type of relationship to a description or a code snippet
- Find mistakes or improvements in a diagram
- Decide, which of the diagrams contains a mistake
- Compare a diagram with a code snippet
- Alter a diagram based on instructions
- Match class diagram components with their descriptions and pictures
- Theoretical questions

The sections with tests will consist of a number of exercises selected from across all the course's topics. The results will be recorded and analyzed later.

### 6.1.5 E-learning techniques

Based on our analysis, we concluded that the most suitable e-learning techniques are microlearning, gamification, adaptive learning and social learning. Our courses will also apply self-directed learning, since a course can be taken at any time and at any place, and blended learning, as it is meant to be a complementary tool to normal classes rather than a stand-alone learning material.

Under microlearning we understand breaking down content in the sections dedicated to explaining the concepts into small pieces.

Gamification offers the largest amount of possible features in an online course out of all the e-learning techniques.

It is important to note that there is a limited number of gamified elements that can be implemented. One of the main constraints on them is that they must not overwhelm or confuse the students. The one-hour format of the courses does not provide the space to use more complicated mechanics associated with gamification, as they might take time to get used to. It was observed that some gamified courses slowed the learners down first before increasing their performance due to the need to adapt to the different format of the study method [kim2015dynamical].

Examples of such techniques are levels, experience points, purchasable items within the course or meaningful quests, which are common instances of gamification. Instead, we want to use storytelling and role-play, which might be even more efficient than artificial awards such as points [8]. We will use the role-play as the main distinction between the 2 versions of the courses, giving the narrator of the instructions and theoretical sections a different personality - one will be fun and warm, the other brief, demanding and professional. We want to further enhance the mood of the courses by selecting appropriate - but applicable - e-learning techniques.

The competitive version of the course will contain a leaderboard and display metrics of other students (more on that later in the same section). We decided not to use a timer for tasks even though it would fit the mood of the course, since it presents hindrance for students with lower base knowledge and encourages rushing through materials.

The playful version of the course will strive to increase the students' emotional engagement by providing rewards for correct answers, while providing "fun failure" for incorrect responses in the form of memes and pictures. Another element to compliment the design are badges awarded for reaching milestones within the course, which are a popular addition to gamified environments - despite not being sufficient to engage students on their own. Lastly, students will be given funny nicknames to attract their attention at the start of the course.

On top of that, we plan to implement visual changes triggered by progress in the courses, mistakes or difficulty level of exercises.

Adaptive learning should allow the course to propose content to the student based on their mistakes in exercises. Feedback should be specific and explain why an answer is incorrect.

When it comes to social learning, collaborations on modeling tasks are not allowed by the experiment format - students will take the courses on their own outside of classes. For this reason, we will restrict the practice to displaying a number of metrics to the users, such as the average score on a given exercise, average completion time or the overall progress of other students in the course. Doing so will apply some social pressure on the students to perform better and potentially trigger their competitive tendencies. This practice is sometimes mentioned as a part of gamification.

Since such practices are sometimes questioned in literature, we will restrict this functionality to the competitive version of the course.

E-learning techniques that are to be implemented:

- microlearning
- gamification (narration, badges...)
- adaptive learning (feedback, content recommendations)
- social learning (performance comparisons, leaderboards - restricted to competitive course)
- self-paced learning
- blended learning

### 6.1.6 Evaluation

Tested students will be divided into 2 groups - each student will only take one of the course versions. That will allow for separate analysis of effects of the different course designs on the students.

The first part of evaluation will consist of a questionnaire with questions about subjective opinions and impressions of the students. The most important questions will concern whether the students thought the course was helpful, whether it improved their understanding of the class diagram and, most importantly, how the e-learning elements impacted their perceptions and behaviors during and after taking the course.

The second, objective part of the evaluation will consist of 2 tests containing theoretical questions and practical exercises about the class diagram. One of the tests will be taken before completing the course, the other one after. They will be used to evaluate the progress of the students. The same tests will be taken by both groups.

The last part of the evaluation will measure the students' behavior - such as time taken to complete activities and the amount of interactions with the system's elements under each course design.

## 6.2 Requirements

From the solution description, we can extract lists of functional and nonfunctional requirements.

### Functional requirements

- Automated exercise/test correction with and instant feedback
- Support for:
  - Gamification- visual feedback, difficulty indicators, badges...
  - Microlearning
  - Adaptive learning - content recommendations based on a student's results, feedback
  - Social learning - displaying other students' performance (anonymized)
- Support for various interactive exercises and learning materials
  - UML code snippets linked to diagrams
  - Full diagram examples with explanation of modeling decisions
  - Drag and drop exercises
  - Matching exercises, multiple choice exercises
  - Theoretical questions
- Students need to provide anonymous feedback on the content
- Tests of students' skills
- Monitoring of students' behavior

### Nonfunctional requirements

- Easy to learn and use
- Open source or free to use
- Capable of storing and retrieving student data
- Scalable and maintainable for future expansion
- Easily extensible (adding new features/content)
- Restricted access - allow access only to students and teachers at FIIT STU
- Content can be accessed anytime, anywhere

### Class diagram of the system

This diagram describes the entire system as we intend it to function. It shows all the major entities that should be found there - users, course and its content, tools for tracking progress and modules for additional features such as adaptive learning.

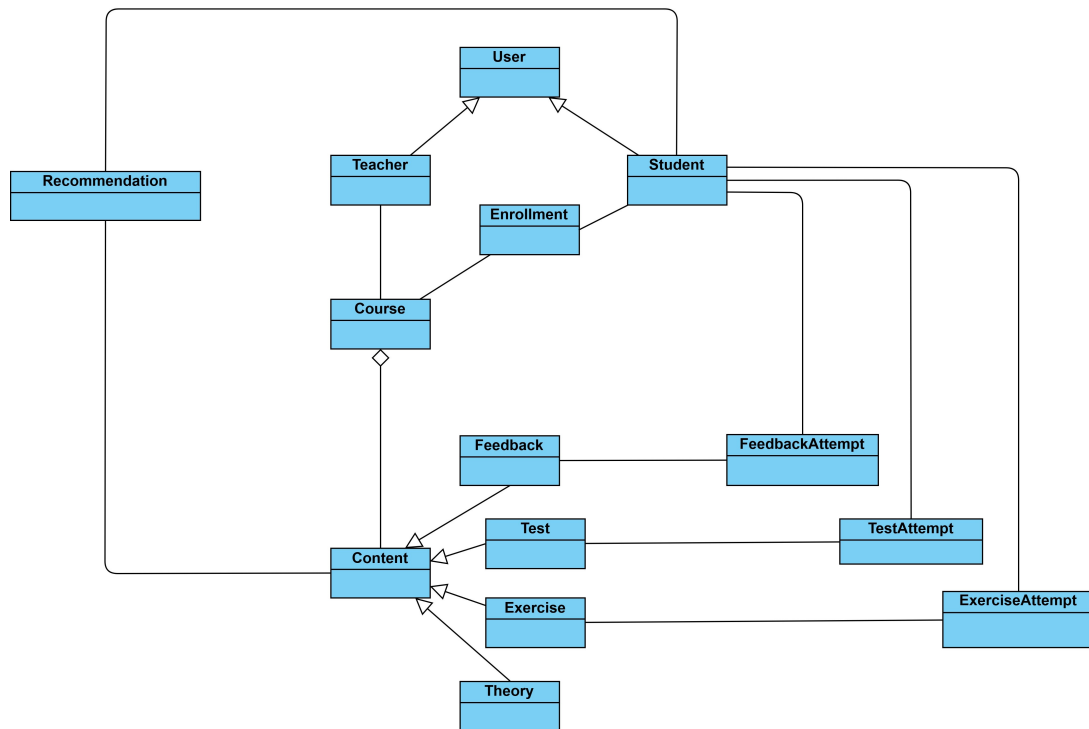


Figure 6.2: Class diagram of the designed system.

Made with: Visual Paradigm, *Visual Paradigm Online*, <https://online.visual-paradigm.com/>, accessed Dec 9, 2025.



### Sequence diagram - entire system

The sequence diagram describes the interaction between all the major entities in the system - teacher, student, user interface, backend. Steps such as authentication were omitted for better clarity. The teacher needs to create courses and their contents. Then the student can enroll in a course and complete it in the flow already described in the activity diagram (see figure 6.1). Lastly, the teacher can review the student's results and make their conclusions and adjustments.

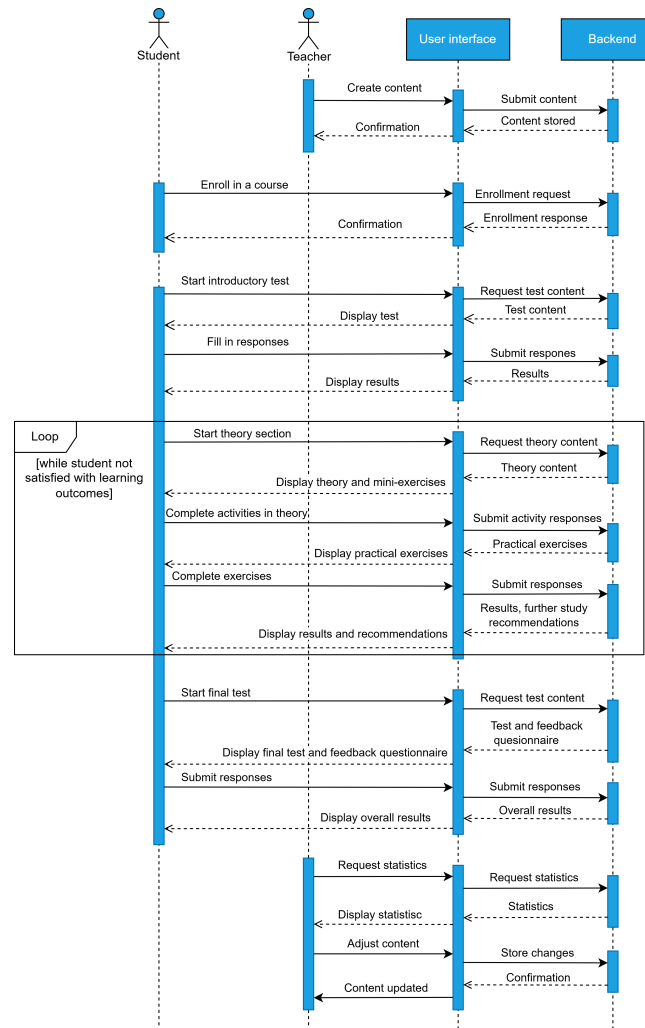


Figure 6.3: Sequence diagram of the whole system's workflow.

Made with: Diagrams.nte, *Diagrams.net*, <https://app.diagrams.net/>, accessed April 30, 2026.



## Chapter 7

# Implementation

We decided to implement our course in Moodle LMS. Moodle satisfies the vast majority of our requirements - both functional and nonfunctional.

It is open source, easy to use, and it is possible to deploy on local servers, allowing us to control the amount of computational resources dedicated to it for future scaling. We can restrict access to a set of users defined by us. It stores information about students' progress, generating statistical reports about their behaviors and results automatically. Content can be added easily, with a lot of plugins developed and maintained by community that introduce features not present in basic Moodle itself.

Moodle supports most of the proposed exercise types out of the box - matching exercises, code snippets, drag and drop, multiple choice, and other. Exercises are corrected automatically, feedback can be provided for each answer in an exercise. Tests and feedback forms are also implemented and ready to use in Moodle.

The biggest hindrance is the absence of content types designed for UML education. However, we found no other LMS that would provide this functionality without the need for a workaround, so that is not a decisive issue. For the first version, we decided to substitute the need for an editor with drag and drop exercises.

### 7.1 Nonfunctional requirements - Moodle<sup>1</sup>

#### Easy to learn and use

Moodle has extensive documentation and multiple tutorials for educators available. The interface for creating content is intuitive and easy to navigate.

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<sup>1</sup>This section was made with the use of Gen AI

It allows generating login credentials for students, meaning the students will receive their credentials via email, log in to the platform and can start completing the course right away. That is desirable, as it allows them to focus solely on learning without the need to set anything up.

### **Open source or free to use**

Moodle is open source, meaning it is free and fully customizable.

### **Capable of storing and retrieving student data**

Moodle stores data about students' quiz and question attempts. The data structure is designed to support adaptive learning practices and detailed statistics. It automatically generates reports on each quiz.

### **Scalable and maintainable for future expansion**

Moodle is designed to host multiple large courses, possibly even functioning as a website for whole educational institutions. We deployed our Moodle site on a private server that belongs to FIIT STU. The allocated resources were selected based on the expected number of users, but additional servers can be dedicated if more users are present.

### **Easily extensible content**

Adding new content or adjusting old one is easy with Moodle, as well as adding additional courses in case we want to create materials for other types of diagrams. Moreover, Moodle supports plugins developed by community that implement various features like new types of exercises or gamification practices.

### **Restricted access**

It is possible to turn off the option to self-enroll into the site. Then we can generate accounts for a list of users from a .csv file, which allows full control over who can access our courses without the need to configure external authentication services such as LDAP or SSO. Should we decide to make our courses public for all internet users to access, self-enrollment can be turned back on.

### **Content accessibility**

As long as the Moodle server is running, students can access the materials.

## 7.2 Course content

### 7.2.1 Theoretical content

We placed the chapter with explanations of concepts of the class diagram right after the introductory assessment quiz and questionnaire. It is implemented as a number of Moodle's book resource<sup>2</sup> instances.

The book resource contains a number of pages with static content. It is structured like an actual book with a table of contents, chapters and subchapters. This way the student is aware and in control of the flow of their progress. It can be exported and printed to a .pdf file for future reference by the students as well.

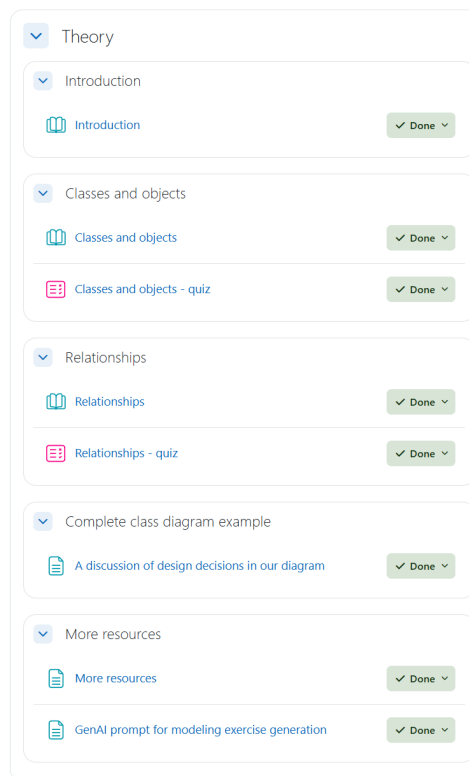


Figure 7.1: Navigation of the theory section

It begins with an introduction that explains the importance of software modeling and what UML is.

The following sections guide the students through the process of building a class diagram -a simple model of an e-shop- one component at a time, explaining each newly

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<sup>2</sup>Moodle, "Book resource", [https://docs.moodle.org/501/en/Book\\_resource](https://docs.moodle.org/501/en/Book_resource), accessed December 13, 2025.

added concept and discussing design choices.

The first book contains chapters about these concepts: entities, classes, objects, types, methods, attributes, access modifiers. It is followed by a short quiz with exercises related to the topics.

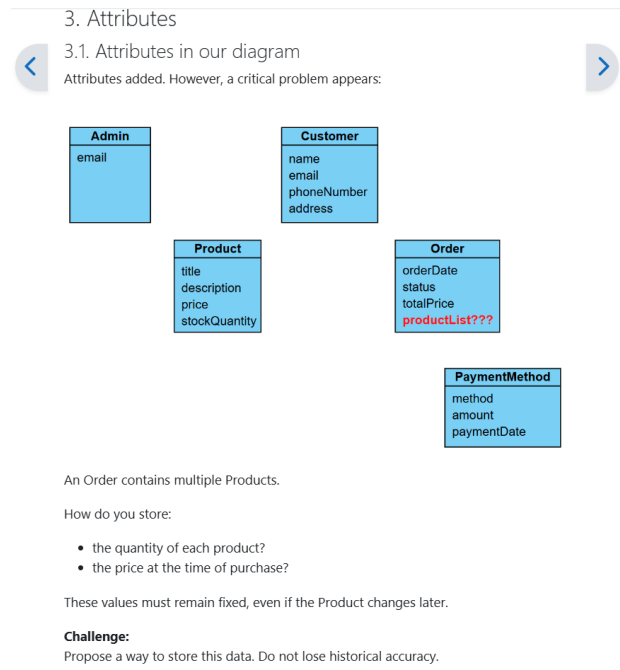


Figure 7.2: Gradually building a diagram while discussing modeling challenges as they arise

The second book talks about relationships - various types of relationships, differences between them, how they can be added to the example diagram, multiplicity and role names. This book has an associated quiz with a few related exercises as well.

The section continues with a picture of the example diagram and a list of modeling decisions the students might question while reading the theory. Those questions include: "Should Admin be a separate class?" or "Why are there no ids, no Product in OrderItem, no getters or setters?"

add a comment about examples of the theory and exercises are in the attached files. When they are attached

### 5. Why are there no ids, no Product in OrderItem, no getters or setters...

Implementation uses them—every class will have an ID.

- **Omission rationale:** UML focuses on conceptual clarity, not code-level detail.

**Conclusion:** Include only what adds value for understanding [relationships](#) and responsibilities.

Figure 7.3: Discussion of design details in the example diagram

The last part of the theoretical section is a short collection of links to more resources about software modeling - a full course, video, article, a link to more examples. There is also a prompt for Gen AI in case students want to easily create exercises of their own.

There were other options to implement this section than the book resource, mainly Moodle's Lesson activity<sup>3</sup>, which provides more space for adaptive learning. However, it cannot be referred to in later stages of the course, which is what made us choose the book instead. Appropriate theoretical sections are linked in feedback to exercises in case a student makes a mistake.

### 7.2.2 Exercises

Moodle allows us to implement the majority of the exercise types described in earlier chapters (see Section 6.1.4). We made 29 exercises in total, most of them focused on simulating modeling diagrams by using Moodle's drag and drop question type<sup>4</sup> and exercises in which the students need to connect diagrams with code. Exercises display hints, reasoning behind correct answers, or links to corresponding theoretical sections after incorrect answers.

All exercise types are:

- Differentiating between relationship types based on description or context
- Find mistakes or room for improvement in a diagram
- Choosing the right relationship arrow directions
- Selecting entities that should be found in a diagram based on a description
- Simulating modeling by dragging components onto empty slots in an image of a diagram
- Theoretical questions - matching access modifiers to their symbols, differentiating between classes and objects...

more examples in  
attached files

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<sup>3</sup>Moodle, "Lesson activity", [https://docs.moodle.org/501/en/Lesson\\_activity](https://docs.moodle.org/501/en/Lesson_activity), accessed October 7, 2025.

<sup>4</sup>Moodle, "Drag and drop onto image question type", [https://docs.moodle.org/502/en/Drag\\_and\\_drop\\_onto\\_image\\_question\\_type](https://docs.moodle.org/502/en/Drag_and_drop_onto_image_question_type), accessed December 19, 2025.

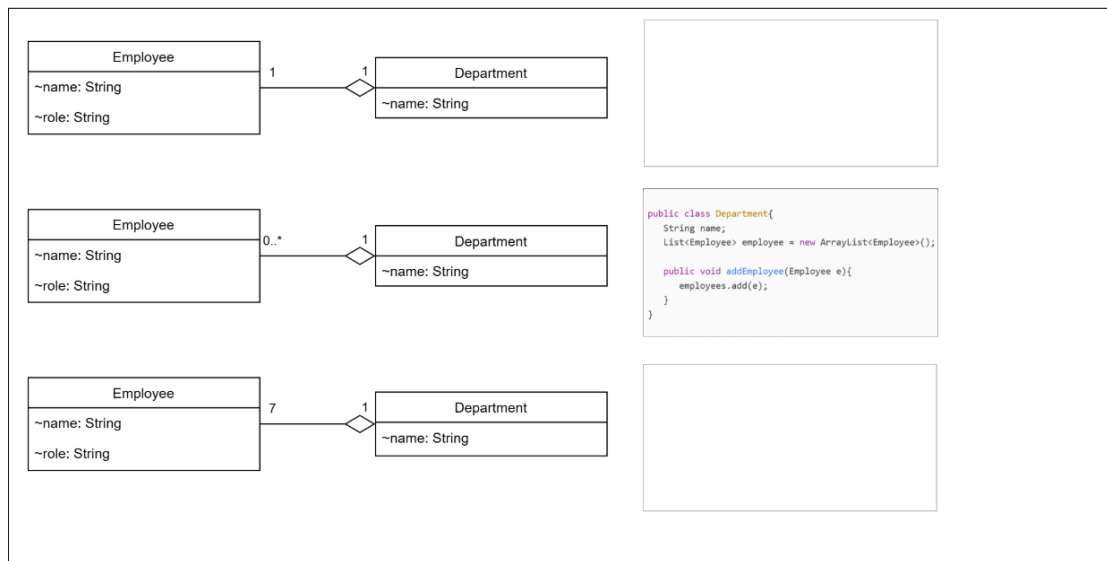


Figure 7.4: Exercise for matching code with corresponding diagrams

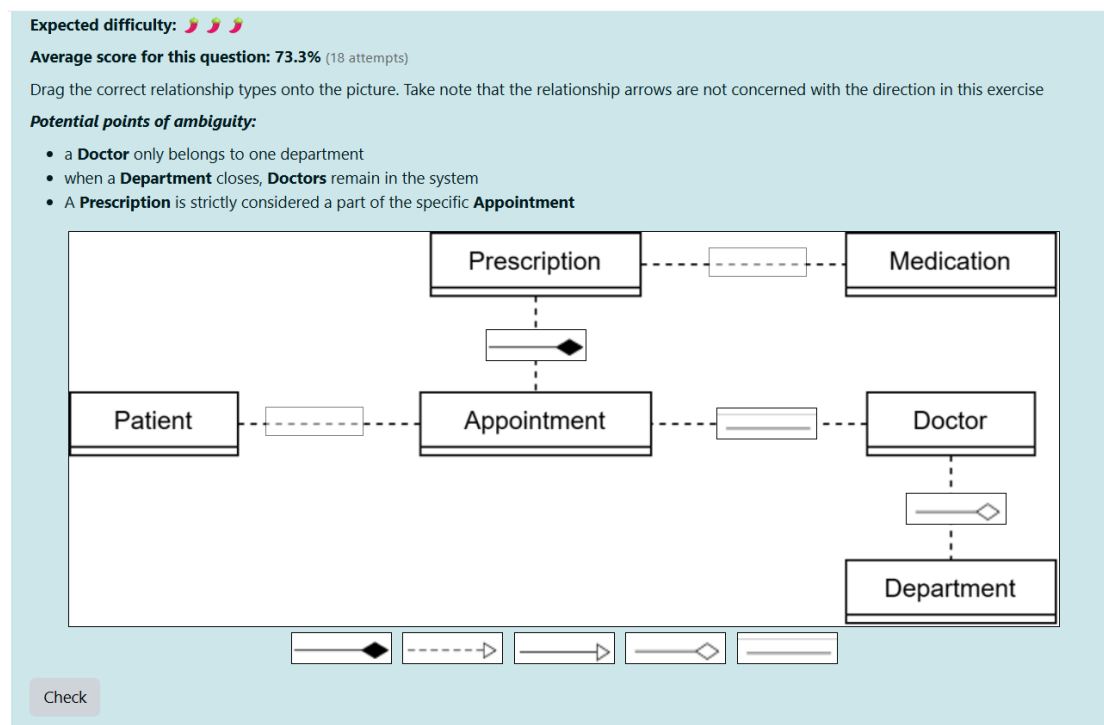


Figure 7.5: Simulating modeling with the drag and drop onto picture activity



Select sensible improvement suggestions

Select one or more:

- ☐ Relationship between Product and OrderItem should be aggregation
- ☒ Relationship between User and Order should be association
- ☐ Order should be in association with User instead of Customer
- ☐ Relationship between Subscriber and Product should be association
- ☒ Relationship between Product and OrderItem should be association
- ☐ Relationship between User and Subscriber should be realization instead of generalization

Check

Figure 7.6: Picking correct improvement suggestions for a flawed diagram

### 7.2.3 Testing students' knowledge

Moodle's quiz activity<sup>5</sup> is a selection of exercises from a defined set. It is possible to have static quizzes that contain the same exercises every time, or select a number of exercises at random from a set.

We used the quiz activity for all the exercise sections. In the quiz settings, the practice sections show instant feedback for each response in each exercise after the student clicks a check button found under each exercise. They are allowed to retry each exercise multiple times, getting hints in between attempts as well. However, retrying a question multiple times reduces the amount of points received for the question.

In the final assessment and introductory assessment sections, the immediate feedback is not present and number of attempts is lower.

### 7.2.4 Statistical reports

Moodle provides statistics regarding individual quizzes, exercises and the overall success of individual students<sup>6</sup>. Reports contain statistics like effective weight or discrimination index mainly useful for adaptive learning practices. We created our own reports aimed at comparing our 2 courses.

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<sup>5</sup>Moodle, "Quiz activity", [https://docs.moodle.org/501/en/Quiz\\_activity](https://docs.moodle.org/501/en/Quiz_activity), accessed October 7, 2025.

<sup>6</sup>Moodle, "Statistics", <https://docs.moodle.org/501/en/Statistics>, accessed December 13, 2025.

## Chapter 7. Implementation

We used the Configurable reports plugin<sup>7</sup> to perform queries on the Moodle database. The plugin allowed for convenient .csv exports of the reports.

Some of the reports were used as dynamically updated statistics for the students - they could view their performance across various categories or compare their performance with other students in the Competitive version of the course . However, most of them were used in the evaluation of the experiments.

refs

this actually belongs to e-learning practices

Manage reports / Intro vs final quiz improvement

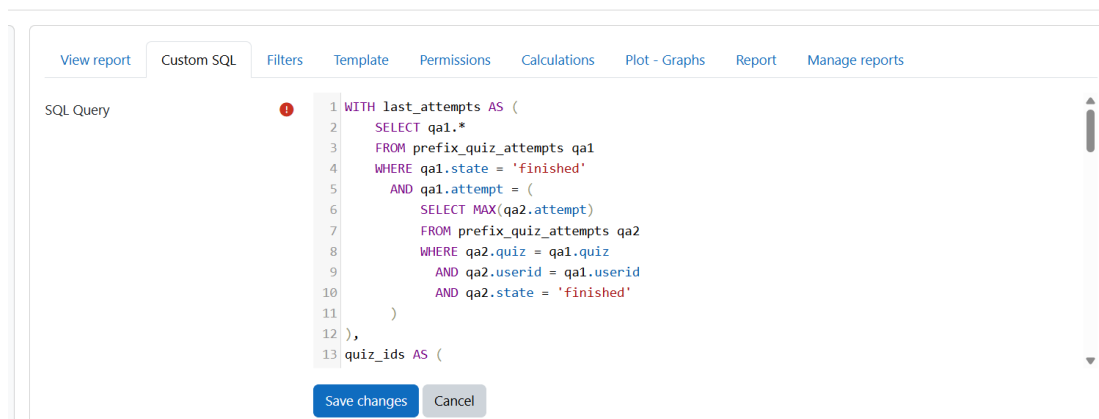


Figure 7.7: Creating a custom report.

| nickname  | intro_score | final_score | improvement |
|-----------|-------------|-------------|-------------|
| subject19 | 69.49       | 92.51       | 23.02       |
| subject13 | 67.82       | 79.67       | 11.85       |
| subject23 | 76.41       | 84.78       | 8.37        |
| subject6  | 83.33       | 90.86       | 7.53        |
| subject34 | 91.92       | 98.66       | 6.74        |
| subject27 | 83.85       | 90.52       | 6.67        |
| subject10 | 82.37       | 86.53       | 4.16        |
| OVERALL   | 82.83       | 87.02       | 1.47        |
| subject14 | 94.87       | 95.67       | 0.80        |
| subject35 | 82.69       | 81.68       | -1.01       |

Showing 1 to 10 of 71 entries

Total record count = 71

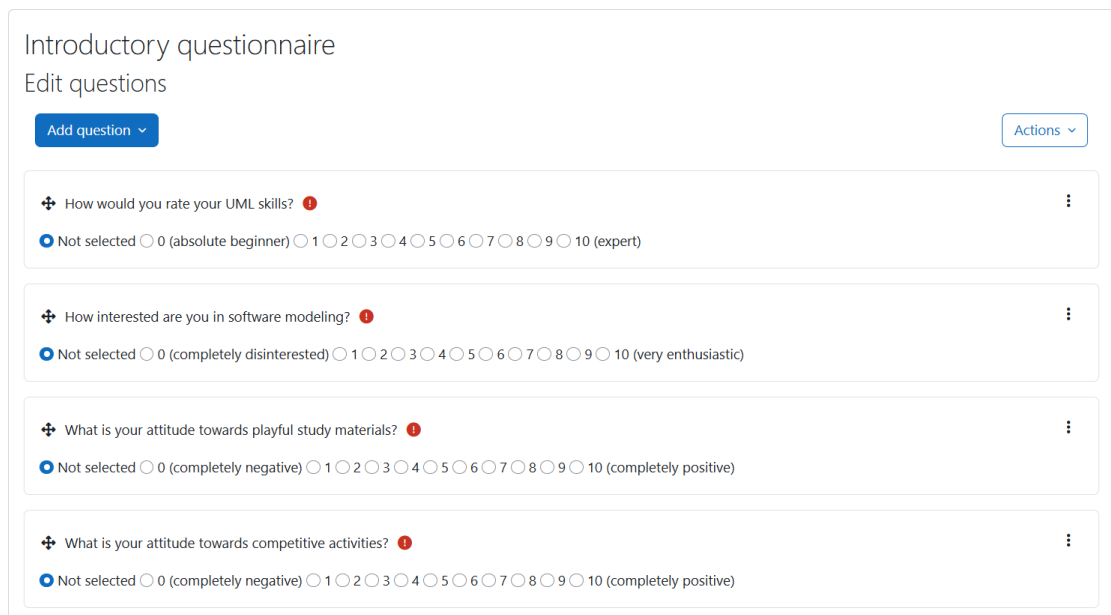
First Previous 1 2 3 4 5 ... 8 Next Last

Figure 7.8: Example of a custom report.

<sup>7</sup>Juan Leyva and Sara Arjona Téllez, "Configurable reports", [https://moodle.org/plugins/block\\_configurable\\_reports](https://moodle.org/plugins/block_configurable_reports), accessed February 13, 2026.

### 7.2.5 Feedback questionnaire

Collecting feedback from participants is one of the most important parts of the whole project. Moodle's feedback activity<sup>8</sup> lets students answer questions about their perception of the course. The responses are anonymous. Both questionnaires were made with the activity. They will be further discussed in the Evaluation sections.



The screenshot displays a Moodle 'Introductory questionnaire' interface. At the top, it says 'Introductory questionnaire' and 'Edit questions'. There are two buttons: 'Add question' (blue) and 'Actions' (blue with a dropdown arrow). Below are four question cards, each with a plus icon, a question text, a red exclamation mark icon, and a vertical ellipsis icon. Each card contains a radio button scale from 0 to 10, with descriptive text at the ends. The first question is 'How would you rate your UML skills?' with a scale from 'Not selected' to '10 (expert)'. The second is 'How interested are you in software modeling?' with a scale from 'Not selected' to '10 (very enthusiastic)'. The third is 'What is your attitude towards playful study materials?' with a scale from 'Not selected' to '10 (completely positive)'. The fourth is 'What is your attitude towards competitive activities?' with a scale from 'Not selected' to '10 (completely positive)'.

Introductory questionnaire  
Edit questions

Add question Actions

⛶ How would you rate your UML skills? !

Not selected 0 (absolute beginner) 1 2 3 4 5 6 7 8 9 10 (expert)

⛶ How interested are you in software modeling? !

Not selected 0 (completely disinterested) 1 2 3 4 5 6 7 8 9 10 (very enthusiastic)

⛶ What is your attitude towards playful study materials? !

Not selected 0 (completely negative) 1 2 3 4 5 6 7 8 9 10 (completely positive)

⛶ What is your attitude towards competitive activities? !

Not selected 0 (completely negative) 1 2 3 4 5 6 7 8 9 10 (completely positive)

Figure 7.9: A sample from feedback questionnaire for students.

<sup>8</sup>Moodle, "Feedback activity", [https://docs.moodle.org/501/en/Feedback\\_activity](https://docs.moodle.org/501/en/Feedback_activity), accessed December 13, 2025.

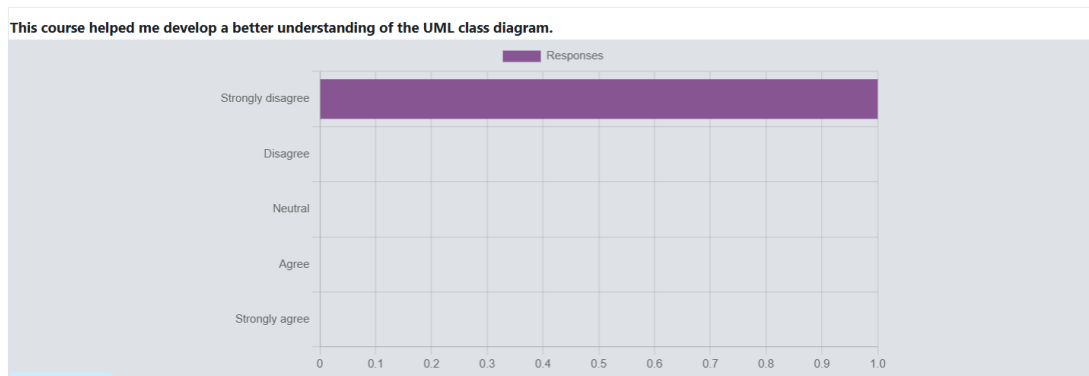


Figure 7.10: Viewing the results of the feedback questionnaire.

### 7.3 E-learning techniques

**Microlearning** - we broke the theoretical section into small chapters, where each chapter contains only 1 concept. There are also exercises placed in between theoretical sections to make the content more engaging.

|                                       |
|---------------------------------------|
| Table of contents                     |
| <a href="#">1. Introduction</a>       |
| <a href="#">2. What is a class?</a>   |
| <a href="#">3. Attributes</a>         |
| <b><a href="#">4. Types</a></b>       |
| <a href="#">5. Methods</a>            |
| <a href="#">6. Access modifiers</a>   |
| <a href="#">7. Classes VS objects</a> |

Figure 7.11: Clear separation of topics for microlearning

**Blended learning** in our courses is implicit. The courses were used as a part of an actual university class, so the participants experienced both e-learning and traditional learning.

**Self-paced learning** - students were allowed to take the course whenever it suited them, and did not have a time limit on finishing it after they started.

**Adaptive learning and instant feedback** - students have access to a report that displays their performance across various categories.

| tag                    | points_earned | points_max | score_percentage | recommendation                               |
|------------------------|---------------|------------|------------------|----------------------------------------------|
| multiplicity           | 10.74         | 12.00      | 89.51            | Very good – minor gaps                       |
| entity_identification  | 8.00          | 10.00      | 80.00            | Very good – minor gaps                       |
| methods_and_attributes | 18.75         | 24.00      | 78.13            | Very good – minor gaps                       |
| code_vs_diagram        | 9.67          | 13.00      | 74.36            | OK – recommend additional practice exercises |
| relationships          | 26.69         | 37.00      | 72.14            | OK – recommend additional practice exercises |
| access_modifiers       | 8.33          | 12.00      | 69.44            | OK – recommend additional practice exercises |

Figure 7.12: A student's performance across various categories.

After answering exercises, students were shown whether the answer was correct, partially correct, or incorrect. In case of incorrect answers, they might be shown hints, links to related theory sections or rationalizations behind the answers.

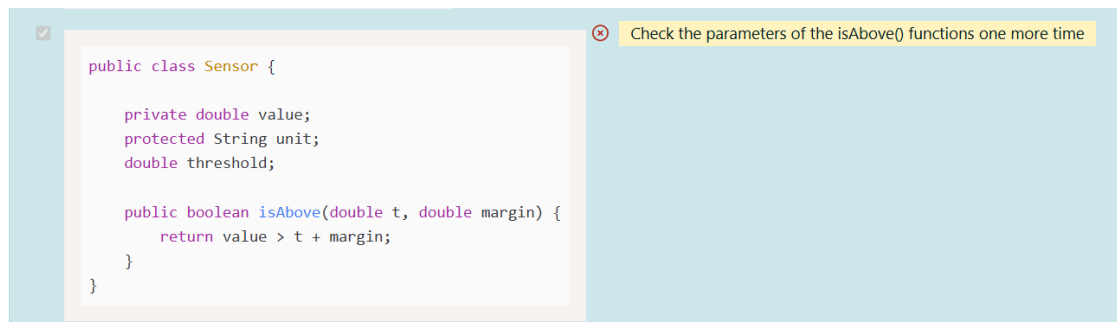
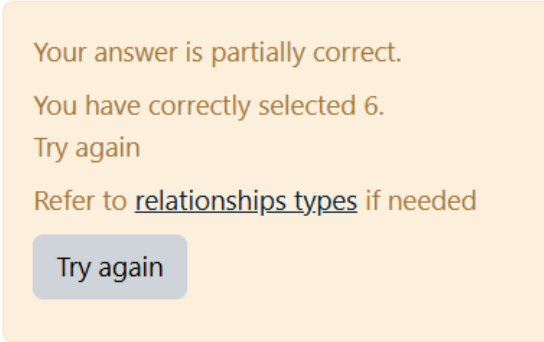


Figure 7.13: An incorrect answer displays a hint to the student.



Your answer is partially correct.  
You have correctly selected 6.  
Try again  
Refer to [relationships types](#) if needed  
[Try again](#)

Figure 7.14: Related theory section is suggested after an incorrect answer.

### 7.3.1 Playful course version

For the playful course version, we implemented several gamification practices:

- Granting rewards for correct answers
- Awarding badges for reaching milestones in the course
- An informal, playful narration of theoretical and informational sections of the course

### 7.3.2 Competitive course version

In the competitive version of the course, the following elements were introduced:

- Leaderboard
- Report that displays the student's performance across categories compared to the course's average
- Each question displays the average score of all the attempts at it across the entire course
- Assertive, strict narration of theoretical and informational sections of the course

why tf did i opt for this, i need to ground this in reasearch that does not state competitive is bad lol

## 7.4 Deviations from design

Several changes were made when compared to the intended design . Some of the proposed content proved either impractical, hard to implement, or there was not enough room in the courses for the ideas to be included. The changes were:

- Course structure change - an additional questionnaire was added at the start of the courses. Originally, it was proposed to only include a questionnaire at the end of the courses
- Providing code snippets along with diagram examples in the theory section proved impractical - the diagram example corresponded to a rather large codebase, and

why tf did i opt for this, i need to ground this in reasearch that does not state competitive is bad lol

ref to design

including even parts of it would have been either insufficient or overwhelming. We only applied comparison of code and diagrams in exercises

- Visual indicators based on the student's current grade. We decided to only include exercise difficulty indicators
- Exercises type Alter a diagram based on instructions was excluded, as there was no way to implement without creating custom activity types in Moodle
- ??

maybe some  
elearning tech-  
niques? Exercise  
recommendations -  
but how to explain  
that i am dumb  
and slow. Any-  
thing else?

### 7.5 Deployment

We were given access to a server owned by FIIT STU for the duration of the experiment (15.3.2026-4.5.2026). The specifics of the server were as follows:

- Operating system: Ubuntu 22.04 LTS <sup>9</sup>
- Number of virtual CPU cores: 2
- Memory size (GB): 4
- Data disk size (GB): 50

The amount of computational resources was requested based on the assumption that no more than 80 users would use the site, and it should handle up to 20 users at a single time. The disk size may seem large for the use case, but Moodle stores a lot of user data.

For the duration of the experiment, domain `fiiit.sk` was kindly provided by Ing. Juraj Vincúr, PhD. which allowed us to set up a secure connection using Let's Encrypt<sup>10</sup>.

Instead of restricting access to the website to only allow users with a FIIT STU Virtual private network (VPN), we disallowed the creation of new accounts. That means we generated accounts to our target students who were then sent credentials via email. We did not implement any additional access restrictions.

---

<sup>9</sup>Canonical Ltd., "Ubuntu Server 22.04 LTS (Jammy Jellyfish) Cloud Image", <https://cloud-images.ubuntu.com/jammy/current/jammy-server-cloudimg-amd64.img>, accessed March 15, 2026.

<sup>10</sup>Let's Encrypt, <https://letsencrypt.org/>, accessed March 15, 2026.





## Chapter 8

# Experiment and Evaluation

### 8.1 Experiment

We tested the created courses on a sample of computer science students at FIIT STU. The students were all in their second year and attending a software modeling class. Each student was given an account on our Moodle site and was enrolled in one of the courses. We decided to assign the course randomly. It is likely that if we let students choose the course variant on their own, only those who enjoy competitive activities would choose the competitive variant. That might give us biased results with overly positive reception of the competitive approach to educational content - which has been demonstrated to be a potentially polarizing factor, as not some people find it more unpleasant than motivating.

ref if possible

Participation in the experiment was voluntary. Students who filled in their assigned course were given bonus points (2) to their software modeling class, and those who scored in top 30% across both courses were given double the credits (4). In total, 35 students completed their course - 14 for the competitive variant, 21 for the playful variant. Some of the participants dropped out, resulting in varying amounts of activity completions throughout the courses.

We collected data describing the participants' behavior, objective performance and subjective perception. The courses were available to students between 7.4.2026 and 24.4.2026.

### 8.2 Data collection

#### Questionnaires

We collected data from 2 questionnaires - 1 taken before and 1 after filling out the

courses. The answers were anonymized, and it is not possible to trace an answer to a specific user. We only knew which course variant the answer belongs to.

Unless stated otherwise, the questions had fixed answer selection on a scale from 0-10, where 0 means strongly disagree and 10 means strongly agree.

The first questionnaire contained questions trying to assess the students' baseline attitudes and self-reported skill level:

- How would you rate your UML skills? (Likert 0-10)
- What is your prior experience with UML? (Multiple choice)
- How interested are you in software modeling? (Likert 0-10)
- What is your attitude towards playful study materials? (Likert 0-10)
- What is your attitude towards competitive activities? (Likert 0-10)

The second questionnaire had significantly more questions, but they were still aimed at the students' attitudes towards e-learning, their satisfaction with the experience and self-reported skill level, including a number of open-ended questions. Full questionnaire is provided in Appendix ??.

- How would you rate your UML skills? (Likert 0-10)
- I would like to encounter more e-learning materials during my studies at FIIT STU. (Likert 0-10)
- This course helped me develop a better understanding of the UML class diagram. (Likert 0-10)
- I would voluntarily take a similar course. (Likert 0-10)
- The course kept me engaged. (Likert 0-10)

### Reports

We collected data regarding the students' objective performance. Names and email addresses were omitted from the reports, but we did keep the user identifiers. There are 2 variants of each mentioned report - one for the competitive course version and one for the playful course. That means though no feature describes which course a data point belongs to, the data are separated by course implicitly. The only exception is `Average_score_per_quiz_per_course`, which contain `courseid` as a feature.

Each question is categorized using tags, allowing us to examine the scores across various topics. We have average scores for each exercise and the overall number of attempts at them. This data was aggregated across all users divided by course variant.

The introductory and final assessment quizzes contain very similar exercises. That makes the difference between their relative scores a reasonable measure of progress of individual students. This data were collected per individual.

Similarly, we examined the performance in the first half of the courses' quizzes com-

ensure this is correct

pared to the second half, and the students' final grades.

- Most difficult question (describes individual exercises)
  - question\_name - name of the exercise
  - quiz\_name - name of the quiz in which the exercise is located
  - tags - categories of the exercise
  - attempts - total number of attempts at the exercise across all users in the course variant
  - avg\_score\_percent - average score in the exercise normalized to a scale of 0-1
  - variability - standard deviation of the score in the exercise
  - difficulty\_label - derived feature, describes an exercise's difficulty based on avg\_score\_percent
- Tag\_difficulty\_cleaner
  - tag - name of a category for exercises
  - average\_score - average score in questions labeled with the associated tag across all users in the course variant
- Average\_score\_per\_quiz\_per\_course
  - course\_id - identifier of the course variant
  - quiz\_name - name of the quiz
  - avg\_score\_percent - average score in the quiz across all users in the course variant
- Final grade across all users
  - finalgrade - the final grade, calculated as a sum of points gained across quizzes in the course. Quizzes had different weights - the final assessment quiz contributed the most significantly
- First half VS second half
  - first\_group\_score - score in the first 3 quizzes (expressed in percents)
  - second\_group\_score - score in the last 2 quizzes (expressed in percents)
  - improvement - calculated as second\_group\_score - first\_group\_score
- Intro vs final quiz improvement
  - intro\_score - normalized score in the Introductory assessment quiz on scale 0-1
  - final\_score - normalized score in the Final assessment quiz on scale 0-1
  - improvement - calculated as final\_score - intro\_score

Full reports are provided in Appendix ??.

### Moodle logs

Moodle stores data about user behavior, such as a user's activity start and end times, interactions with the system, or number of attempts for activities. We gain insights into how many times they viewed theory sections and how many times they interacted with

ensure this is correct and anonymized correctly

quizzes (most of those interactions are reattempts at exercises). We collected this data for each user separately.

Measuring time taken to go through the course is unreliable, since the material is self-paced and students could take breaks in between course activities. For this reason, we approximated the time spent by only including quiz attempts. The data are still noisy, because the students could still take breaks in the middle of a quiz. But that is less likely and left us with only a handful of outliers.

The most important features of the resulting datasets were:

- Interactions
  - interaction\_type - name of the interaction type (theory/quiz/feedback)
  - activity\_name - name of the activity (Introductory assessment quiz, Classes and objects theory page...)
  - interactions - number of interactions with an activity
- activity\_time\_sums
  - total\_time\_seconds - total time spent on quizzes in seconds
  - total\_attempts - number of attempts at quizzes in total (5 at minimum)

Full reports are provided in Appendix ??.

ensure this is correct

### 8.3 Data preparation

#### Questionnaire

#### Reports and logs

We combined the logs and reports from the previous section into a single table for better clarity and processing. This was only applied to reports that contained data for individual students - data about exercises, categories and questionnaire answers were not included.

link to appendix

-Derived features: -theory/quiz interactions ratio -restructure interactions so that they are in buckets -theory clicks

-Data bucketing -duration cat -(interactions)

-cleaning -time was capped at 2:15 hrs per user -removed feedback interactions

The resulting dataset contained features describing each student's performance, behavior and derived metrics. The resulting features were:

- id - user id for the participants' identification
- finalgrade -
- improvement -

add derived features when made up, add explanations to the features

- intro\_score -
- final\_score -
- duration\_mins -
- total\_attempts -
- quiz -
- theory - number of interactions with the theory pages
- !!!!!!!!!!!

### Exercises

The data needed very little cleaning

There was one exercise with very low scores that made the categories it belonged to appear more difficult than they were. The exercise was removed from the categories, but kept for overall aggregate measurements.

## 8.4 Result analysis

-disregarded reports -research questions (maybe in summary, or a separate section)

## 8.5 Limitations

pekne rozpisat

any other issues  
with data?

- Anonymized questionnaires
- some data gathered is tricky to interpret - time, interactions
- size and time of the course - not allow for some of the more advanced activities in e-learning, cannot overload the students
- baseline knowledge - students were already familiar with the class diagram
- setting - impossibility of direct group activities
- exercises - would be better to have an automated assessment tool and use draw.io and some diagram-code conversion tool instead of adapting general purpose exercise types, but outside the scope of 1 DP project
- moodle - what is difficult to do there?
- activities restricted to "no intervention from teachers" - no larger or more complex exercises, only automated correction
- visual animations - we do not have that now, maybe leaderboard like in kahoot and such, transitions, changes based on performance...

## 8.6 Summary



## **Chapter 9**

# **Conclusion**

### **9.1 Summary**

### **9.2 Future Work**





# Resumé

## Úvod

Strojové učenie je dôležitá oblasť umelej inteligencie...



# List of Abbreviations

**FIIT STU** Faculty of Informatics and Information Technologies STU in Bratislava

**Gen AI** Generative artificial intelligence

**LMS** Learning Management Systems

**UML** Unified Modeling Language

**VPN** Virtual private network



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## References

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# Declaration about the use of Generative Artificial Intelligence

OpenAI (2026), ChatGPT, Section 6.3, consulting the design of diagrams

OpenAI (2026), ChatGPT, Chapter 7, Moodle and server debugging

OpenAI (2026), ChatGPT, Sections 7.3, creation of content displayed in courses (generation of text, consulting exercise design)

OpenAI (2026), ChatGPT, Section 8.3, generating code for data analysis, language Python

this should refer to  
labels - navigation

## Appendix A

# Description of Digital Submission

**Thesis Evidence Number:**

FIIT-182905-116308

**Name of submitted archive:**

DP\_MatejSevcik.zip

actual file names,  
make the big  
pretty file struc-  
ture

when they are  
cleaned up

when they are  
cleaned up, wait  
until you have the  
reports finished

The digital part included in the thesis contains the following files:

-backup file without students -generico filter template backup -configurable reports  
backup -analysis jupyter files + pdf exports -



## Appendix B

# Technical Documentation

### B.1 Installation Manual for Local Development

The installation guide assumes the use of Windows 10 or 11 operating systems.

1. Moodle provides installer packages for Windows that contain everything that is needed to run the application server locally. Download the latest stable release [from the official website](#) (the version name should not end with dev. E.g. - instead of downloading *Moodle 5.3dev*, download *Moodle 5.2*)
2. Create a directory in a location where you want your local Moodle instance to reside
3. Extract the installer package into the newly created directory
4. Navigate to the extracted directory and run `Start Moodle.exe`
5. Visit <http://localhost/> and follow displayed instructions to complete the installation
6. (Optional) If you want to recreate the courses used in this study:
  - (a) Log in as your admin user
  - (b) Install plugins according to instructions (Section B.3)
  - (c) Navigate to **Site administration** -> **Courses** -> **Restore course** ([Course restore documentation](#))
  - (d) Drop one of the attached backups (`course_backups/competitive_backup_no_users.mbz` or `course_backups/playful_backup_no_users.mbz`) into the drop zone and click **Restore**. Do so for both attached courses individually



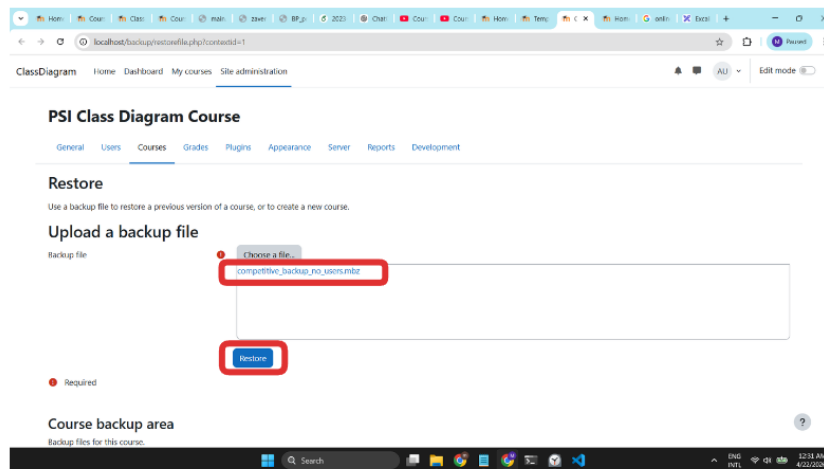


Figure B.1: Restore a course from a backup in Moodle

- (e) All the restore settings can be left at defaults. Pay care to select **Restore as new course** instead of **Restore into a course**

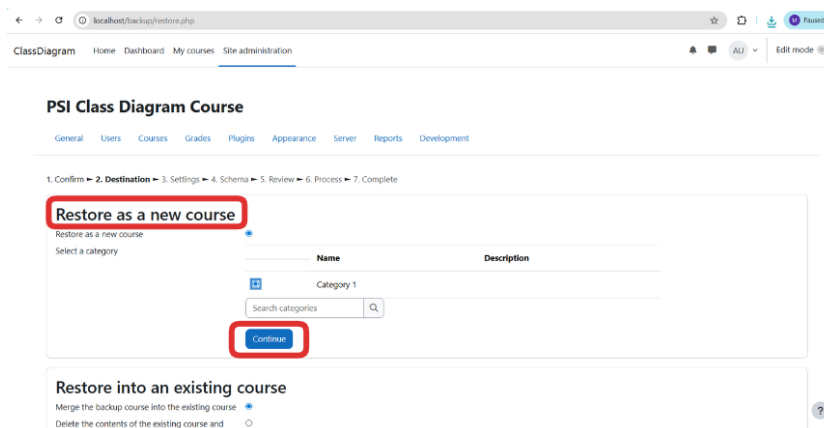


Figure B.2: Restore a backup as a new course

- (f) Once the restore is pending, you need to run the cron task. On a production server, it runs automatically, but locally it must be executed manually from the command line. Open a terminal and run (the entire text is one command):
- ```
"<path-to-moodle>\server\php\php.exe"  
"<path-to-moodle>\server\moodle\admin\cli\cron.php"
```

## B.2 Installation Manual for Production Environment

1. Obtain access to a Linux server (Recommended: Ubuntu 24.04.4 LTS)
2. Set up the server according to [instructions from Moodle documentation](#)
3. (Optional) To recreate the courses used in this study, follow instructions from the previous section (item 6)

## B.3 Plugin Installation

Plugins are not restored using course backups. Courses from this study work without plugins, but some of the e-learning features are not available without them. They are used for getting live statistics and reports. To recreate them, follow these instructions - they are the same for both production and local environments:

1. Download plugins [Generico filter](#) and [Configurable reports](#)
2. Navigate to **Site administration -> Plugins -> Install plugins** ([Plugin installation documentation](#)). Drop the downloaded zip files into the displayed field and click **Install plugin from the ZIP file**

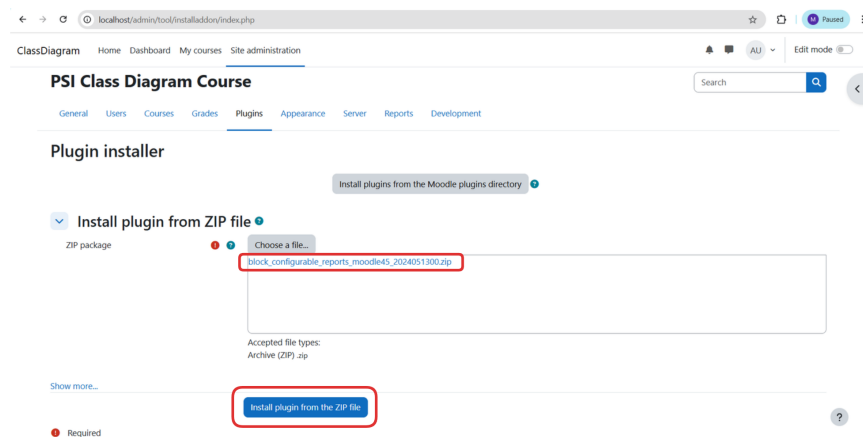


Figure B.3: Installing a plugin from a ZIP file in Moodle

### 3. Configurable reports

- (a) Navigate to <course-home> side panel and go to **Configurable Reports -> Manage reports**

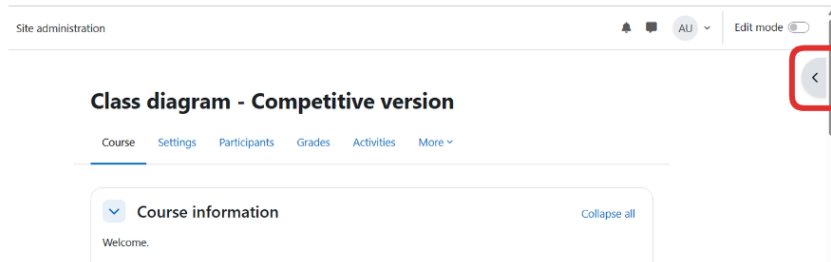


Figure B.4: Recreating Configurable reports in Moodle, step 1

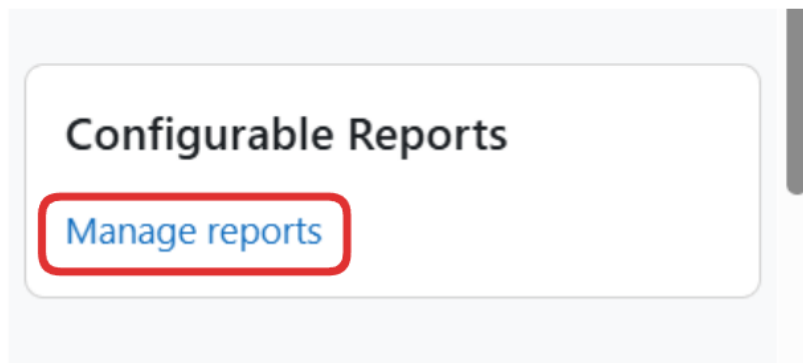


Figure B.5: Recreating Configurable reports in Moodle, step 2

- (b) Drag each report backup (`configurable_reports/competitive_course_backups/interactions.`) to the drop area in the corresponding course

## Appendix B. Technical Documentation

ClassDiagram Home Dashboard My courses Site administration

version

Add report

Import report from repository

Categories

Report

Import report

File

Choose a file...

interactions.xml

Import report

Required

Figure B.6: Recreating Configurable reports in Moodle, step 3

### 4. Generico filter

- (a) Go to **Site administration** -> **Plugins** -> **Filters** -> **Category: Generico** -> **Templates** and click on one of the blank filters

Filters / Generico / Templates

PSI Class Diagram Course

General Users Courses Grades Plugins Appearance Server Reports Development

Generico Templates Admin

| Name         | Template key | Version | Description                                                                          |
|--------------|--------------|---------|--------------------------------------------------------------------------------------|
| Welcome User | welcomeuser  | 1.0.1   | This is a simple demonstration of how to display the current Users name and picture. |
| ?            |              |         |                                                                                      |
| 3            |              |         |                                                                                      |
| 4            |              |         |                                                                                      |

Figure B.7: Recreating a Generico filter in Moodle

- (b) copy values from `generico/generico_question_avg.txt` to their respective fields
- (c) Each time the filter is used in the courses, the id in the filter is hardcoded. That means it needs to be manually updated in each instance (e.g. questions

in the Competitive course contain {GENERIC0:type=question\_avg,qid=507}  
in their descriptions - the id will need to be updated)

# Appendix C

## Work Schedule

### C.1 Work Schedule for DP1

|            |                                                                 |
|------------|-----------------------------------------------------------------|
| week 1     | planning and outlining the work, researching the topic          |
| week 2-3   | writing about UML and the class diagram                         |
| week 4-5   | research and writing about e-learning                           |
| week 6-8   | researching the applications of e-learning to the class diagram |
| week 9-10  | writing about e-learning's applications to the class diagram    |
| week 11-12 | proposing our approach to the rest of the project               |

**Statement on the execution of the plan:** The purpose of DP1 was to research the topic of the project. At the beginning of the semester, exploration of UML and e-learning was carried out. We described the basic concepts of the class diagram. The subsequent topics were drawn out, however. We got delayed at research about e-learning and its applications on UML, and had to compensate for the lost time during the exam period.

## C.2 Work Schedule for DP2

|           |                                                          |
|-----------|----------------------------------------------------------|
| week 1    | research about various LMS                               |
| week 2-3  | familiarizing with Moodle LMS                            |
| week 4-6  | implementing the theoretical sections of the course      |
| week 7-11 | implementing the concepts of the exercises in the course |
| week 12   | writing the design and implementation chapters           |

**Statement on the execution of the plan:** The goal for DP2 was to implement the course. The intentions were not completely fulfilled, as we still have a number of tasks left, namely creating more exercises, implementing gamification and social learning. We did create a draft of the course as well as drafts for all the exercise types we want to use.

### C.3 Work Schedule for DP3

|            |                                                         |
|------------|---------------------------------------------------------|
| week 1-2   | implementing gamification and social learning practices |
| week 3-4   | finishing creating additional exercises                 |
| week 5     | deployment of the course on a private FIIT STU server   |
| week 6-10  | performing experiments on students and evaluating them  |
| week 10-12 | finish writing the text for the thesis                  |